

IT3081 STATISTICAL MODELLING · GROUP CONSULTANCY REPORT

Customer Repurchase and Value Prediction for a UK Online Retailer

Predicting which customers will return, what they are worth, and how to spend the retention budget

Client scenario: a UK online giftware wholesaler · Dataset: UCI Online Retail II (2009–2011)

Prepared 01 October 2026 · Task 11 (expert validation) pending

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Executive Summary

Client question. A UK online giftware wholesaler wants to know which customers will buy again, how much they are worth, and where to spend its retention budget.

Data. The public UCI *Online Retail II* dataset: 1,067,371 invoice lines from December 2009 to December 2011. After cleaning, 1,021,252 lines remain. They are summarised into a table of **5,253 identifiable customers**, with behaviour measured before 9 September 2011 and outcomes measured in the 90 days after. Customer-level findings apply only to account-holding customers: 22.8% of raw lines (13.6% of revenue) have no customer ID, and that missingness is not random.

What we found

Area	Key result
Customer base (Task 3)	Revenue is highly concentrated: the top 10% of customers bring in 62.7%. Only 43.4% of customers bought again in the next 90 days. Demand peaks in November at about 2.2 times a typical month
Statistical inference (Task 4)	International customers' typical order value is 1.68 times UK customers' (95% CI 1.55–1.82) and far more variable. Q4-acquired customers repurchase 9.6 points more often (95% CI 6.7–12.5), a small effect. Early spending rate differs only slightly by acquisition cohort ($\omega^2 = 0.022$)
Prediction (Task 5)	Under a Brier-score rule declared in advance and applied over four rolling-origin periods, LASSO logistic regression is selected (Ridge is statistically indistinguishable). Locked and evaluated on a later period, it ranks customers well (AUC 0.79; top 10% repurchase at 2.1 times the average) and beats a “most recent first” rule, but its probability level is miscalibrated across seasons. Log-OLS is selected for spend
Advanced methods (Tasks 7–9)	PCA adds nothing to the customer table. Product-level SVD reveals buying themes but barely improves prediction. Bayesian pooling gives stable estimates for 26 small-country markets. Two years of data are too few for a reliable seasonal forecast: the fitted model lost to a naive one

The central caveat. The models predict who will buy under the conditions seen historically. They cannot tell whether a retention offer changes that. Depending on the offer's unknown effect, the profitable share of customers to contact ranges from 4% to 82%. No £ figure in this report is a deployable profit forecast.

What we recommend (Task 12; provisional until the expert consultation in Task 11 is completed)

1. Do not fund blanket retention discounts yet.
2. Run a four-month randomised pilot (Task 6), launched outside the Q4 peak, with repurchase as the success measure and pre-agreed stop/go rules.
3. Build the weekly early-warning system (Task 10) in stages: dashboard first, campaigns only once the pilot has measured the offer's effect.
4. Protect the top accounts with service contact rather than discounts.
5. Plan Q4 from the calendar, starting in June–July, not from a forecasting model.

Status of the deliverables. Tasks 1–10 and 12 are complete in this report. **Task 11 (industry expert validation) has not yet been carried out**, so it appears as a consultation plan with empty evidence templates. The expert-feedback column of every recommendation is marked *Pending*.

About this report

Team

Member	Student ID	Responsible for
A	IT24102584	Task 3: data cleaning and descriptive analysis
B	IT24103124	Tasks 4, 6, 7: statistical inference, experimental design, PCA
C	IT24102616	Tasks 5, 8: predictive modelling, Bayesian methods
D	IT24103989	Tasks 1, 2, 9, 10, 11: problem framing, literature review, time series, innovation proposal, expert validation
All		Task 12: final recommendations

Reproducibility. Every number in this report is produced by code in the project repository: Python notebooks in `notebooks/python/` (00 data cleaning, 01 descriptive analysis, 03 predictive modelling, 04 PCA, 05 Bayesian methods, 07 experimental-design power analysis) and R scripts in `notebooks/r/` (02 statistical inference, 06 time series). Notebook 00 must be run first. The data split that prevents leakage (features before 9 September 2011, outcomes after it) is implemented once, in `src/features.py` and `R/features.R`, and used everywhere.

Use of generative AI. *[To be confirmed and completed by the group before submission.]* An AI coding assistant (Claude, by Anthropic, used through Claude Code) was used to help review and implement parts of the analysis code, check results for errors, and draft sections of this report. The group reviewed all AI-assisted output, reran every analysis, and remains responsible for the accuracy and originality of the work, as required by the brief's academic integrity section.

Placeholder assumptions. Margin (30%), offer cost (£10) and offer effect (10% of expected spend) are **assumptions**, not client data. They are labelled wherever they are used and are to be replaced after the expert consultation.

Task 1 – Understanding the Industry Problem

1.1 Industry Background

The consultancy operates within **UK online retail and business-to-business (B2B) gift wholesale**, a sector that sits at the intersection of e-commerce and bulk trade. Our client data source is the **Online Retail II** dataset (UCI Machine Learning Repository), which records approximately **1.07 million invoice lines** generated by a UK-based online gift wholesaler between **December 2009 and December 2011**.

Unlike a typical consumer e-commerce retailer, this business does not sell primarily to individual end consumers. Its customer base consists largely of **independent gift shops and small retailers** who purchase decorative items, novelty gifts, and seasonal stock in bulk for resale. This distinction matters: transaction volumes per order are higher than in consumer retail (median basket value of approximately **£275**), and purchasing behaviour is driven by *stock-replenishment* cycles rather than impulse buying.

Geographically, the business is overwhelmingly UK-concentrated: **85.3% of total revenue** and approximately **91% of customers** are based in the United Kingdom. The remaining revenue is distributed across roughly 40 other countries, with Ireland (EIRE), the Netherlands, Germany, France, and Australia representing the largest international markets.

1.2 Organizational Context

The client is best characterised as a **small-to-mid-sized online gift wholesaler** operating on comparatively thin margins relative to the size of individual transactions. Two structural features of the business are critical to understanding the problem:

- **Seasonality.** Revenue is strongly concentrated around Q4, with **November revenue roughly 2.2 times** that of a typical off-peak month, driven by retailers stocking up ahead of the Christmas selling season. This pattern repeats across both years of available data (November 2010 and November 2011 both show comparable peaks).
- **Operational rhythm.** The business takes **almost no orders on Saturdays** (0.1% of revenue), reflecting either a B2B ordering pattern (retail buyers ordering during their own business hours) or an operational choice by the wholesaler. Order volume peaks between 10:00 and 15:00 on weekdays.

1.3 Description of the Business Problem

The central organizational problem is that **revenue is highly concentrated among a small number of customers, and the business has no systematic method for identifying which customers are likely to stop purchasing**.

Quantitatively, from our descriptive analysis of the customer-level data (5,253 identified customers after data cleaning):

- The **top 1% of customers generate 30.8%** of total identifiable revenue.
- The **top 10% of customers generate 62.7%** of total identifiable revenue.
- The **top 20% of customers generate 76.4%** of total identifiable revenue.

This is a textbook Pareto (80/20) distribution of customer value, and it means the business's income is **structurally exposed to the loss of a small number of high-value buyers**. Compounding this exposure, our analysis of a 90-day forward-looking window shows that only **43.4% of customers made a repeat purchase – 56.6% did not**. Non-repurchase is not a rare edge case for this business – it affects more than half of the customer base – yet there is currently no mechanism to flag at-risk customers before they are lost, nor any evidence-based way to decide how much retention spend a given customer relationship justifies.

1.4 Importance of Solving the Problem

The economic case for addressing this problem rests on well-established **retention economics**: acquiring a new B2B wholesale customer is more costly than retaining an existing one, because acquisition requires marketing spend, onboarding, and relationship- building with an unproven buyer, whereas an existing customer already has a demonstrated purchase history and trust relationship. Given the revenue concentration described above, losing even a small number of top-decile customers would have a **disproportionate impact** on annual revenue relative to losing an equivalent number of low-value customers.

Additionally, **undifferentiated retention spending is inefficient**. Contacting every customer with a discount offer wastes margin on customers who would have repurchased regardless (encouraging discount-seeking behaviour without incremental benefit), while failing to prioritise the customers whose departure would be most damaging. A statistically grounded targeting approach directly addresses this inefficiency.

1.5 Stakeholders Affected

Stakeholder	Interest in the Problem
Marketing team	Needs to know <i>which</i> customers to target for retention and <i>what</i> offer to extend, within a limited budget.
Finance team	Needs to protect profit margin and requires a defensible, quantified justification for retention spend.
Operations / inventory management	Needs reliable demand forecasts to plan stock and staffing ahead of the concentrated Q4 selling period.
Wholesale customers (small retailers)	Depend on consistent stock availability and fair, non-discriminatory commercial treatment; are indirectly affected by how the business allocates its attention and offers.

1.6 Business Decisions Influenced by the Problem

1. **Which customers receive retention offers**, and at what intensity (e.g. discount email vs. reminder-only contact vs. no contact).
2. **How the marketing budget is allocated** across the customer base – concentrated on high-value at-risk customers, or spread broadly.
3. **How Q4 inventory and staffing are planned**, given the predictable seasonal demand spike and the operational absence of Saturday trading.

1.7 Consultancy Objectives

Our engagement is structured around four objectives, each of which maps to a specific analytical task in this report:

1. **Describe** purchasing behaviour and quantify revenue concentration (Task 3).
2. **Test statistically** whether meaningful customer segments (e.g. UK vs. international, Q4-acquired vs. not) differ in value and loyalty (Task 4).
3. **Build and compare predictive models** for repurchase probability and future customer spend (Task 5), evaluated not just on accuracy but on calibration and expected business profit.
4. **Recommend a practical, profit-aware retention strategy** for management, validated against real-world constraints identified through literature and industry expert feedback (Tasks 10–12).

1.8 Dataset Selection and Justification

Dataset. Chen, D. (2012). *Online Retail II* [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5CG6D>

The dataset is the full transaction log of a UK-based, non-store online gift wholesaler. It was released by the same research group that first analysed its predecessor for RFM-based customer segmentation (Chen, Sain & Guo, 2012), so the business context is documented in peer-reviewed literature rather than assumed.

Brief requirement (§6)	How Online Retail II meets it
Directly addresses the business problem	It records <i>who</i> bought <i>what</i> , <i>when</i> , and for <i>how much</i> , for every order over two years. Repurchase and future spend – the two outcomes the client cares about – can be observed directly by splitting the timeline at a cutoff date (9 Sep 2011) and looking at the following 90 days. This mirrors the calibration/holdout design used throughout the customer-base-analysis literature (Schmittlein, Morrison & Colombo, 1987; Fader, Hardie & Lee, 2005b).
Sufficient observations	1,067,371 raw invoice lines; after cleaning, 1,021,252 lines, 46,919 invoices, 4,739 products, 43 countries. The customer-level modelling table has 5,253 customers , of whom 43.4% repurchased – a balanced outcome with roughly 2,280 positive cases, far above the usual events-per-variable guidance for a 9-predictor logistic model.
Variables for descriptive, inferential and predictive analysis	The 8 raw columns support RFM feature construction (Recency, Frequency, Monetary), tenure, basket value, product breadth, cancellation rate and country. These give continuous, binary and categorical variables – enough for comparisons of means, proportions and variances, ANOVA (Task 4), classification and regression (Task 5), PCA (Task 7), hierarchical models across 41 countries (Task 8), and a two-year daily/weekly revenue series (Task 9).
Publicly accessible	Free download from the UCI repository under a CC BY 4.0 licence; no registration or data-sharing agreement needed, so every result in this report is reproducible.
Properly referenced	Cited above with its DOI, alongside the original peer-reviewed study that introduced the data (Chen et al., 2012).

Why this dataset over alternatives. Most public churn datasets (e.g. telecom subscription data) are *contractual*: churn is observed when a contract ends. Our client is *non-contractual* – a wholesale customer never “cancels”, they simply stop ordering – and this setting is both harder and less studied in B2B contexts (Tamaddoni Jahromi, Stakhovych & Ewing, 2014; Gattermann-Itschert & Thonemann, 2022). Online Retail II is one of very few public datasets that capture that setting at the transaction level, and it has been used in recent peer-reviewed repurchase-prediction research on online retailing (Chou et al., 2022).

Known limitations (stated here so they are not discovered in the viva):

- **No cost or margin data.** Profit-based evaluation (Task 5) must use assumed margin and offer-cost figures, to be validated with the industry expert (Task 11).
 - **No marketing-contact or customer-demographic data.** Gattermann-Itschert & Thonemann (2022) found that contact-related features (e.g. recency of last sales-rep contact) matter beyond RFM in B2B churn; we cannot use them.
 - **22% of lines have no Customer ID**, and this missingness is not random (notebook 01), so the customer-level models under-represent some order types.
 - **Only two years of history**, which limits seasonal time-series modelling (Task 9).
 - **No record of whether any retention offer was ever made**, so causal effects of an offer cannot be estimated from this data alone – the reason Task 6 proposes an experiment.
-

Task 2 – Research Landscape

2.1 Structure of the Review

The review is organised around the decisions this project had to make. For each approach we took, it identifies the research that supports it; for each approach we considered and dropped, it identifies the research that explains why. It covers:

1. **Current industry practice** in customer retention (§2.2).
2. **Existing statistical approaches** to repurchase and churn prediction, with their strengths and weaknesses (§2.3).
3. **Recent research findings**, 2021 onwards (§2.4).
4. A **critical comparison table** of the core domain studies (§2.5).
5. **Research gaps** (§2.6).
6. A **decision-by-decision justification** of our analytical approach (§2.7).

All 45 references in §2.9 (44 papers plus the dataset) were checked against the Crossref and DataCite DOI registries (authors, year, journal, volume, pages, DOI). Claims attributed to each paper are limited to what its published abstract or well-established content states.

2.2 Current Industry Practices

RFM segmentation is the default starting point. Firms without a data-science function typically segment customers by Recency, Frequency and Monetary value and target segments by rule. Chen, Sain and Guo (2012) – the original study on this dataset – did exactly this, clustering customers with k-means on RFM variables and profiling segments with decision trees, explicitly to help a small online retailer that “lacks the necessary knowledge and expertise” adopt customer-centric marketing. Fader, Hardie and Lee (2005a) show that RFM is more than a heuristic: under a formal probability model, recency and frequency together determine a customer’s expected future value, which they visualise as “iso-value curves”.

Managerial heuristics decide who gets contacted. In practice, retention campaigns are often targeted by simple rules such as “contact customers who have not ordered recently”. Tamaddoni Jahromi et al. (2014) compare model-driven targeting against such managerial heuristics in a non-contractual B2B setting and find the model-driven approach outperforms them.

Churn scoring followed by targeting the highest-risk customers is the dominant model-based practice (Neslin et al., 2006; Gattermann-Itschert & Thonemann, 2022). The recent field study by Gattermann-Itschert and Thonemann (2022), run at a B2B wholesaler, found that contacting the customers with the highest predicted churn probabilities significantly reduced churn compared with random targeting.

2.3 Existing Statistical Approaches

Family	Representative studies	Strengths	Weaknesses
Probabilistic “buy-till-you-die” (BTYD) models (Pareto/NBD, BG/NBD, BG/BB)	Schmittlein et al. (1987); Fader et al. (2005a, 2005b); Abe (2009)	Built for non-contractual settings; parsimonious (need only recency, frequency, tenure); strong behavioural story; produce customer lifetime value directly.	Use few inputs, so cannot exploit other features (cancellations, product breadth, country); rely on distributional assumptions (Poisson purchasing, memoryless dropout).

Family	Representative studies	Strengths	Weaknesses
Regression / GLM classifiers (logistic regression, penalised regression)	Neslin et al. (2006); De Caigny, Coussement & De Bock (2018); Chou et al. (2022)	Interpretable coefficients (odds ratios); well-calibrated probabilities; penalised versions (LASSO, elastic net) handle correlated predictors and select variables (Tibshirani, 1996; Zou & Hastie, 2005).	Assume a linear relationship with the log-odds; may miss interactions that tree-based models capture.
Machine-learning classifiers (SVM, boosting, random forests, neural networks)	Tamaddoni Jahromi et al. (2014) (boosting); Gordini & Veglio (2017) (SVM); Gattermann-Itschert & Thonemann (2022) (random forests)	Flexible; can capture non-linearities and interactions; often competitive or best on ranking metrics.	Harder to explain to managers; probabilities often poorly calibrated without post-processing (Niculescu-Mizil & Caruana, 2005); need tuning.
Profit-driven evaluation	Verbeke et al. (2012); Verbraken, Verbeke & Baesens (2013)	Choose the model – and the fraction of customers to contact – that maximises campaign profit, not a statistical score.	Require cost and benefit parameters that firms often cannot state precisely.
Causal / uplift targeting	Ascarza (2018); Lemmens & Gupta (2020); Devriendt, Berrevoets & Verbeke (2021); Haupt & Lessmann (2022)	Target customers whose behaviour the offer will <i>change</i> , which is what actually generates profit.	Require data from randomised experiments (treated vs. control customers), which most firms – including our client – do not have.

2.4 Recent Research Findings (2021 onwards)

- **Hybrid models beat either approach alone, and LASSO keeps them interpretable.** Chou et al. (2022), using a large online-retailing dataset, found that feeding BG/BB predictions into a high-dimensional LASSO regression significantly outperformed both BG/BB alone and LASSO alone, and that the LASSO-based model “provides interpretability” by identifying the most influential features among about 100 predictors.
- **Churn prediction works in non-contractual B2B wholesale – and so does acting on it.** Gattermann-Itschert and Thonemann (2022) validated a churn model in a live field experiment at a B2B wholesaler, and found that features beyond standard RFM (such as recency of the last sales-rep contact) mattered.
- **Predicting risk is not the same as predicting the effect of an offer.** Devriendt et al. (2021) argue that uplift models, which estimate the *incremental* effect of a retention action, are the better paradigm for retention decisions, and introduce a profit-based measure (maximum profit uplift) to compare them with standard churn models. Haupt and Lessmann (2022) extend profit-optimal targeting to the case where the cost of a marketing action depends on the customer’s response (e.g. a discount only costs money if used).
- **Simple benchmarks remain hard to beat in retail forecasting.** Fildes, Ma and Kolassa (2022) review retail forecasting and highlight “the dimensionality problem of too many variables and too little data”. The M5 competition (Makridakis, Spiliotis & Assimakopoulos, 2022), based on Walmart sales, is the largest recent benchmark of retail forecasting methods.

2.5 Critical Comparison of Core Studies

Study	Setting / data	Method	Evaluation	Strengths	Weaknesses / gap	Used in this project
Chen, Sain & Guo (2012)	UK online gift retailer (this dataset’s predecessor)	RFM + k-means + decision tree	Segment profiles (descriptive)	Same business; accessible to non-experts	Descriptive only – no prediction of future behaviour, no validation	Motivates our RFM features; we go further by <i>predicting</i> repurchase and spend

Study	Setting / data	Method	Evaluation	Strengths	Weaknesses / gap	Used in this project
Schmittlein, Morrison & Colombo (1987)	Theory of non-contractual customer bases	Pareto/NBD	Fit to holdout period	Defines the “is this customer still active?” problem when churn is unobserved	Only recency and frequency; strong distributional assumptions	Justifies defining repurchase in a fixed future window rather than a churn date
Fader, Hardie & Lee (2005a)	CDNOW online music retailer	Pareto/NBD + gamma-gamma spend model, iso-value curves	Holdout tests	Formally links RFM to lifetime value	Spend and purchase processes assumed independent	Justifies RFM as predictors of future value, and modelling purchase and spend as two parts
Neslin et al. (2006)	Telecom churn tournament, many teams	Logit, trees, others	Top-decile lift, Gini, validation on data 3 months later	Large, fair comparison of methods; models keep their accuracy over time	Contractual telecom setting	Justifies top-decile lift as a metric and a single calibration/validation time split
Verbeke et al. (2012)	Telecom churn, multiple datasets	Many classifiers, profit-based metric	Maximum profit from contacting the optimal fraction of customers	Moves evaluation from accuracy to money	Contractual telecom setting	Basis for our expected-profit curve and “% of customers to contact”
Verbraken, Verbeke & Baesens (2013)	Customer churn	Expected maximum profit (EMP) criterion	EMP	Formal cost-benefit framework; guides campaign size	Needs cost and benefit parameters	Same; our profit parameters are placeholders to be validated in Task 11
Tamaddon Jahromi, Stakhovych & Ewing (2014)	Non-contractual B2B	Several data-mining models; boosting best	Prediction of true churners, campaign profit	Closest setting to our client; beats managerial heuristics	Boosting is a black box	Justifies our “contact the most recent customers” heuristic benchmark
Gordini & Veglio (2017)	B2B e-commerce	SVM with AUC-based parameter selection	AUC	B2B e-commerce setting	SVM gives no interpretable coefficients or calibrated probabilities	Considered and dropped (see §2.7)
De Caigny, Coussement & De Bock (2018)	Customer churn, multiple datasets	Hybrid of logistic regression and decision trees	Predictive performance and comprehensibility	Shows logistic regression remains a strong, comprehensible base	Hybrid model is more complex to deploy	Supports logistic regression as our base model
Chou et al. (2022)	Large online retailer	BG/BB + LASSO	Repurchase prediction accuracy	Recent; online retail; LASSO chosen for interpretability	Did not evaluate calibration or profit	Justifies LASSO; BG/BB integration listed as future work

Study	Setting / data	Method	Evaluation	Strengths	Weaknesses / gap	Used in this project
Gattermann-Itschert & Thonemann (2022)	Non-contractual B2B wholesaler, field experiment	Random forest	Churn reduction in a live experiment vs random targeting	Only study here with a real field test in our setting	Random forest less interpretable; needs CRM contact data we lack	Confirms RFM features matter in B2B wholesale; supports proposed experiment (Task 6)
Ascarza (2018)	Two field experiments	Machine learning + randomised trials	Retention achieved by different targeting rules	Shows highest-risk customers are <i>not necessarily</i> the best to target	Needs experimental data	States the main limitation of our approach; motivates Task 6
Lemmens & Gupta (2020)	Retention campaigns	Causal machine learning	Campaign profit	Optimises profit directly	Needs experimental data	Future direction once an experiment has run
Devriendt, Berrevoets & Verbeke (2021)	Customer churn	Uplift vs predictive models	Maximum profit uplift	Recent, profit-based comparison of the two paradigms	Needs treatment/control data	Same
Haupt & Lessmann (2022)	Marketing campaigns	Causal hurdle model	Campaign profit	Handles costs that depend on the customer's response	Needs experimental data	Informs how offer cost should be modelled once real costs are known

What the comparison shows. The studies split along two lines. *First*, interpretable statistical models (logistic, LASSO, BTYD) versus flexible machine learning (SVM, boosting, random forests): the machine-learning studies often report the best ranking performance, but the interpretable models are competitive (De Caigny et al., 2018; Chou et al., 2022) and are the only ones whose outputs a marketing team can act on and audit. *Second*, risk-based versus effect-based targeting: almost all prediction studies rank customers by risk, while the experimental studies (Ascarza, 2018; Devriendt et al., 2021) show that risk is not the same as responsiveness to an offer. Only one study in our setting, Gattermann-Itschert and Thonemann (2022), closes the loop with a real field test.

2.6 Research Gaps

- 1. Evaluation is usually accuracy-only.** Most churn and repurchase studies report AUC or accuracy. Few also check whether predicted probabilities are *calibrated* – whether a “30% chance” really means 30% – even though any profit calculation multiplies by those probabilities (Van Calster et al., 2019). Profit-based evaluation exists (Verbeke et al., 2012; Verbraken et al., 2013) but is rarely combined with calibration and interpretability in the same study.
- 2. Non-contractual B2B settings are under-studied** compared with contractual consumer churn (Tamaddoni Jahromi et al., 2014; Gattermann-Itschert & Thonemann, 2022).
- 3. Future spend is often ignored.** Most churn models predict *whether* a customer leaves, not *how much* they would spend if they stay, even though value, not headcount, drives retention budgets (Fader et al., 2005a).
- 4. Few studies compare several variable-selection methods on the same business data** – stepwise, best subset and penalised regression are usually used one at a time, although the statistical literature shows none of them dominates in general (Hastie, Tibshirani & Tibshirani, 2020).

How this project addresses them. We evaluate every repurchase model on discrimination (AUC), calibration (Brier score, calibration curve), top-decile lift *and* expected profit; we study a non-contractual B2B wholesaler;

we model both repurchase and future spend and combine them into expected value; and we compare seven variable-selection and penalisation strategies on the same split.

2.7 Justification for Our Analytical Approach

Each row is a decision taken in the project, the research that supports it, and what we chose

not

to do as a result.

Problem set-up and data (Tasks 1, 3)

Decision	Research support	What we dropped, and why
Define the target as “repurchase within 90 days after a cutoff”, with features from before the cutoff	In non-contractual settings a customer’s departure is never observed, so activity must be inferred from purchases in a defined window (Schmittlein et al., 1987); calibration/holdout splits are the standard validation design (Fader et al., 2005b), and churn models retain accuracy when applied to data three months later (Neslin et al., 2006)	A “churn date” target – it does not exist for a wholesaler whose customers never formally cancel
Use RFM plus tenure, basket value, product breadth, cancellation rate, country and acquisition quarter as features	RFM formally determines future customer value (Fader et al., 2005a); RFM is the established segmentation basis for this very business (Chen et al., 2012); RFM features are among the most important in B2B wholesale churn (Gattermann-Itschert & Thonemann, 2022)	Using RFM alone – the extra features are cheap to build and the B2B literature shows non-RFM features add information
Test whether missing Customer IDs are random before deciding how to treat them	Rubin (1976) defines when missingness can be ignored; Little (1988) gives the formal test of “missing completely at random”	Silently dropping or imputing the 22% of rows without an ID – our tests showed the missingness is <i>not</i> random

Statistical inference (Task 4)

Decision	Research support	What we dropped, and why
Welch’s t-test for comparing means	Welch’s test controls false positives better when variances differ and loses little when they are equal, so it should be the default (Delacre, Lakens & Leys, 2017)	Student’s t-test – it assumes equal variances, which Test 3 shows do not hold
Brown-Forsythe (median-centred Levene) test for comparing variances	Robust to non-normal data (Brown & Forsythe, 1974)	The classical F-test of variances – very sensitive to non-normality, and customer-level average order values are heavily right-skewed
Welch ANOVA with Games-Howell post-hoc tests	Classical ANOVA can be severely biased under unequal variances; Welch’s F is recommended by default (Delacre, Leys, Mora & Lakens, 2019)	Classical one-way ANOVA
Report effect sizes (Cohen’s d, Cramér’s V, variance ratio, omega-squared) with every p-value	With very large samples p-values shrink towards zero even for trivial effects (Lin, Lucas & Shmueli, 2013); effect size, not p, shows practical importance (Sullivan & Feinn, 2012)	Reporting p-values alone – with 5,253 customers, Test 2 is “significant” yet its effect is small ($V = 0.09$)

Predictive modelling (Task 5)

Decision	Research support	What we dropped, and why
Logistic regression as the base repurchase model	Logistic regression is a strong, comprehensible base in churn prediction (Neslin et al., 2006; De Caigny et al., 2018); its odds ratios can be explained to management	SVM (Gordini & Veglio, 2017) and boosting (Tamaddoni Jahromi et al., 2014) – no interpretable coefficients, and their probabilities typically need recalibration before use in a profit rule (Niculescu-Mizil & Caruana, 2005)
Penalised regression (Ridge, LASSO, elastic net) as candidates; LASSO selected under a pre-declared Brier rule over rolling-origin development periods	LASSO shrinks and selects variables (Tibshirani, 1996); elastic net handles groups of correlated predictors (Zou & Hastie, 2005); LASSO was chosen for interpretability in recent online-retail repurchase prediction (Chou et al., 2022)	The unpenalised model – $\log_Monetary$ has $VIF \approx 200$ because $Monetary \approx Frequency \times AvgBasketValue$, which makes its coefficients unstable
Fit Ridge as a comparison	Ridge regression was introduced for exactly this problem of correlated (“non-orthogonal”) predictors (Hoerl & Kennard, 1970): it stabilises their coefficients by shrinkage	Nothing: Ridge is a full candidate. It keeps every feature (no sparsity) and is statistically indistinguishable from LASSO under the selection rule
Fit forward/backward stepwise and best subset as comparisons, not as the final model	Neither best subset nor LASSO dominates in general; LASSO does better when the signal is noisy, best subset when it is strong (Hastie et al., 2020)	Stepwise as the final model – it biases coefficients, ignores the multiple testing done during selection, and can pick unstable models (Whittingham et al., 2006; Smith, 2018)
Select on the Brier score; also report log-loss, AUC, average precision, calibration intercept/slope and reliability diagrams, not accuracy alone	The Brier score measures the accuracy of probability forecasts (Brier, 1950); poorly calibrated models can mislead decisions (Van Calster et al., 2019)	Accuracy or AUC alone – they ignore whether the probabilities can be trusted, and our profit rule multiplies by them
Report top-decile lift, a retrospective profit curve on realised spend, and a prospective decision rule on predicted spend	Top-decile lift is a standard churn metric (Neslin et al., 2006); choose the model and campaign size that maximise profit (Verbeke et al., 2012; Verbraken et al., 2013)	Choosing the model on AUC – all seven variants tie within 0.001 AUC; and presenting realised-spend profit as deployable – it uses spend not known at decision time
Benchmark against a “most recent customers first” heuristic	Model-driven targeting should be shown to beat managerial heuristics (Tamaddoni Jahromi et al., 2014)	—
Model spend on repurchasers only and multiply by P(repurchase) (a two-part model)	Future value splits naturally into whether a customer buys and how much they spend (Fader et al., 2005a)	A single spend model on all customers – 57% of outcomes are exactly zero, which a Gamma GLM cannot model
Compare OLS on log(spend) with a Gamma GLM (log link), and apply Duan’s smearing to the OLS predictions	Which of the two is better depends on the data’s skewness and variance pattern, so both should be tested (Manning & Mullahy, 2001); back-transforming a log model without a correction underestimates the mean, and the smearing estimate corrects it without assuming normality (Duan, 1983)	Picking one spend model on theory alone
Do not use a probabilistic BTYD model as the main model	BTYD models are parsimonious but use only recency and frequency (Chou et al., 2022)	Dropped as the <i>main</i> model because it cannot use our other features; Chou et al. (2022) show BTYD predictions can be added <i>into</i> a LASSO model – listed as future work

Critical evaluations (Tasks 6–9)

Decision	Research support	What we dropped, and why
Task 6: Recommend a randomised experiment before rolling out targeted offers	Randomised controlled experiments are the best design for establishing cause and effect, but need attention to power, sample size and variance reduction (Kohavi et al., 2009); the highest-risk customers are not necessarily the ones an offer helps (Ascarza, 2018); profit comes from the offer's incremental effect (Lemmens & Gupta, 2020; Devriendt et al., 2021)	Treating the repurchase model as proof that offers work – it predicts who will return, not who an offer will change
Task 7: Do not use PCA on the customer feature table; on the sparse customer × product matrix use Truncated SVD (a PCA-related method that skips centring) on a binary bought / not-bought matrix	PCA reduces dimensionality while losing as little information as possible, and helps interpretation when data are large (Jolliffe & Cadima, 2016); in regression, low-variance components can still be the most predictive, so dropping them can lose signal (Jolliffe, 1982)	PCA on the 9-feature table – no real compression, coefficients lose their business meaning, and LASSO already handles the collinearity; SVD on raw quantities – its leading components were each driven by a single bulk buyer
Task 8: Report Naive Bayes as a baseline, not a recommendation	Naive Bayes can classify well even when its independence assumption is violated (Domingos & Pazzani, 1997), but pushes probabilities towards 0 and 1 (Niculescu-Mizil & Caruana, 2005)	Using Naive Bayes probabilities in the profit rule – our Brier score (0.212 vs 0.175) confirms the calibration problem
Task 8: Hierarchical Bayesian model to pool information across countries	Shrinking many small-group estimates towards the overall mean improves them (Efron & Morris, 1975); hierarchical Bayes is established in customer-base analysis (Abe, 2009); weakly informative priors are recommended for the between-group standard deviation (Gelman, 2006)	Separate per-country estimates – 26 of 41 countries have fewer than 10 customers
Task 9: Weekly aggregation, auto.arima, and comparison with a naive forecast	Automatic ARIMA selection as implemented in R's <code>forecast</code> package (Hyndman & Khandakar, 2008); retail forecasting evidence stresses benchmarking against simple methods (Fildes et al., 2022; Makridakis et al., 2022)	Deploying a seasonal forecast – two seasonal cycles are not enough, and our ARIMA did not beat the naive benchmark

2.8 Meeting the Minimum Requirements

Requirement	Required	This review
Peer-reviewed research papers	≥ 15	44 (43 journal articles + 1 peer-reviewed conference paper, ICML 2005)
Journal papers	≥ 10	43
Published within the last five years (2021 or later)	≥ 5	6: Devriendt et al. (2021); Chou et al. (2022); Gattermann-Itscher & Thonemann (2022); Haupt & Lessmann (2022); Fildes et al. (2022); Makridakis et al. (2022)

The dataset (Chen, 2012) is cited separately and is not counted as a paper.

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Task 3 – Dataset Understanding and Descriptive Analysis

Owner: Member A. Numbers come from `notebooks/python/00_data_cleaning.ipynb` (cleaning and data-quality summary) and `01_descriptive_analysis.ipynb` (missing values, descriptive statistics, business insights).

3.1 Dataset description

Item	Value
Source	UCI Machine Learning Repository, <i>Online Retail II</i> (Chen, 2012), publicly available
Business	A UK-based online retailer of all-occasion giftware; many of its customers are wholesalers
Period	1 December 2009 to 9 December 2011 (two Excel sheets, one per year)
Raw size	1,067,371 invoice lines × 8 columns
Unit of observation	One product line on one invoice: not one customer, and not one order

Raw variables

Variable	Type	Meaning
Invoice	Categorical (ID)	Invoice number; a leading “C” marks a cancellation
StockCode	Categorical (ID)	Product code
Description	Text	Product name
Quantity	Numeric, discrete	Units on the line; negative for returns and cancellations
InvoiceDate	Date-time	When the transaction happened
Price	Numeric, continuous	Unit price (£)
Customer ID	Categorical (ID)	Customer account; often missing
Country	Categorical, nominal	Customer’s country (about 40 values)

The raw file has only two genuinely numeric variables, so it is not suitable for modelling as it stands. The analysis dataset is a **customer-level table** built from it.

Customer-level analysis table: 5,253 customers × 12 columns. Every feature is computed only from transactions **before 9 September 2011** (the cutoff). Both targets are computed only from the **90 days after** it. This split prevents information from the future leaking into the predictors.

Variable	Definition
Recency	Days from the customer’s last purchase to the cutoff
Frequency	Number of distinct (non-cancelled) invoices before the cutoff
Monetary	Total spend (£) before the cutoff
TenureDays	Days from first purchase to the cutoff
AvgBasketValue	Monetary / Frequency : the customer’s average order value (£)
DistinctProducts	Number of different products bought
CancellationRate	Cancelled invoices ÷ all invoices
IsUK	1 if the customer’s most frequent country is the UK
AcquiredInQ4	1 if the first purchase was in October–December

Variable	Definition
Repurchase (target)	1 if the customer bought anything in the 90 days after the cutoff
FutureSpend (target)	Spend (£) in those 90 days; 0 for non-repurchasers

3.2 Data-quality assessment

The cleaning steps, applied in this order, and the rows each removed (notebook 00, data-quality summary):

Step	Rows removed	Rows remaining	Reason
Raw invoice lines		1,067,371	
Non-product codes, base list (postage, bank charges, fees, manual entries)	5,617	1,061,754	Not product sales; postage would inflate overseas customers' spend
Price ≤ 0	6,182	1,055,572	Free items, warehouse stock notes ("short", "mixed"), bad-debt write-offs such as -£53,594. This step also removes all 3,457 negative-quantity lines that are not cancellations, since every one has price £0
Exact duplicate rows	34,038	1,021,534	The same line entered twice would double-count revenue (one invoice was overstated by 13%)
Extended non-product list (samples, adjustments, gift vouchers, test items, charity commission)	277	1,021,257	Found during descriptive analysis; every removed row was listed and checked before deletion
Mistaken orders reversed within minutes (invoices 581483/C581484, 541431/C541433, 556444)	5	1,021,252	Data-entry errors, e.g. 80,995 units (£168k) cancelled 12 minutes later. Both halves are removed: keeping only the sale would invent a £168k customer, while keeping both would invent a 50% cancellation rate

Kept, but flagged or handled:

- **Cancellations:** 17,912 clean lines (1.75%) are kept, flagged `IsCancellation`, and used to compute each customer's `CancellationRate`. Cancellation behaviour is itself a retention signal.
- **Very large genuine orders** (e.g. 19,152 mugs at £0.10 for a Danish wholesaler) are kept. They have no matching cancellation, a plausible bulk price, and a repeat-buying history: they are real wholesale behaviour. They are handled with log transforms and influence checks rather than deleted. **The rule applied was to delete errors, not extremes.**

3.3 Missing-value analysis

	Lines	Share
Raw lines with no <code>Customer ID</code>	243,007	22.77%
Clean lines with no <code>Customer ID</code>	227,089	22.24%
Net revenue on clean lines with no ID	£2,568,806	13.6% of net revenue

`Customer ID` is the only variable with missing values. The missingness was **tested rather than assumed random**:

- Rows without an ID differ systematically from rows with one: smaller quantities, higher unit prices, more non-UK countries (Hong Kong and Bermuda are 100% missing), and different times of day. Every univariate test is significant after Holm correction.

- A logistic model predicts whether a line is missing its ID with **AUC 0.80**, so the IDs are **not missing completely at random** (Rubin, 1976; Little, 1988). The most plausible explanation is a separate order channel that is not linked to customer accounts.

Treatment. An identifier cannot be imputed: a “median customer” is meaningless, and guessing would invent purchase histories. Missing-ID lines are therefore **kept for revenue totals, trends and country analysis**, and **excluded from the customer-level table**, because a prediction has to belong to a customer. **Consequence: every customer-level conclusion in Tasks 4, 5 and 8 applies to identifiable, account-holding customers only.** These customers under-represent the smaller, higher-priced, non-UK orders that make up the missing-ID lines. The customer table itself has no missing values.

3.4 Outlier detection

- **Line level:** 99% of lines are below £179, but the largest is £15,818 (9,360 units). The raw maximum quantity of 80,995 was an error and was removed (3.2).
- **Customer level:** customer spend has a skewness of 24.3. The mean (£2,580) is 3.3 times the median (£776), the 99th percentile is £24,400, and the maximum is £456,780. On the log scale the distribution is roughly symmetric, which is why Tasks 4 and 5 work with logs.
- **Influence in models:** a single repurchaser (2 orders and £97 of history, then £53k of spend) has a Cook’s distance of 8.7 in the Gamma spend model (Task 5). The models report influence diagnostics rather than deleting such customers.

3.5 Descriptive statistics (customer table, n = 5,253)

Variable	Mean	Median	Min	Max
Recency (days)	205.7	162	0	646
Frequency (orders)	5.7	3	1	284
Monetary (£)	2,580	776	1.55	456,780
Distinct products	74.5	41	1	2,178
Tenure (days)	429.9	472	0	646
Average order value (£)	362.7	275.0	1.55	14,845
UK customers	91.3%			
Q4-acquired	32.9%			
Repurchased in next 90 days	43.4%			

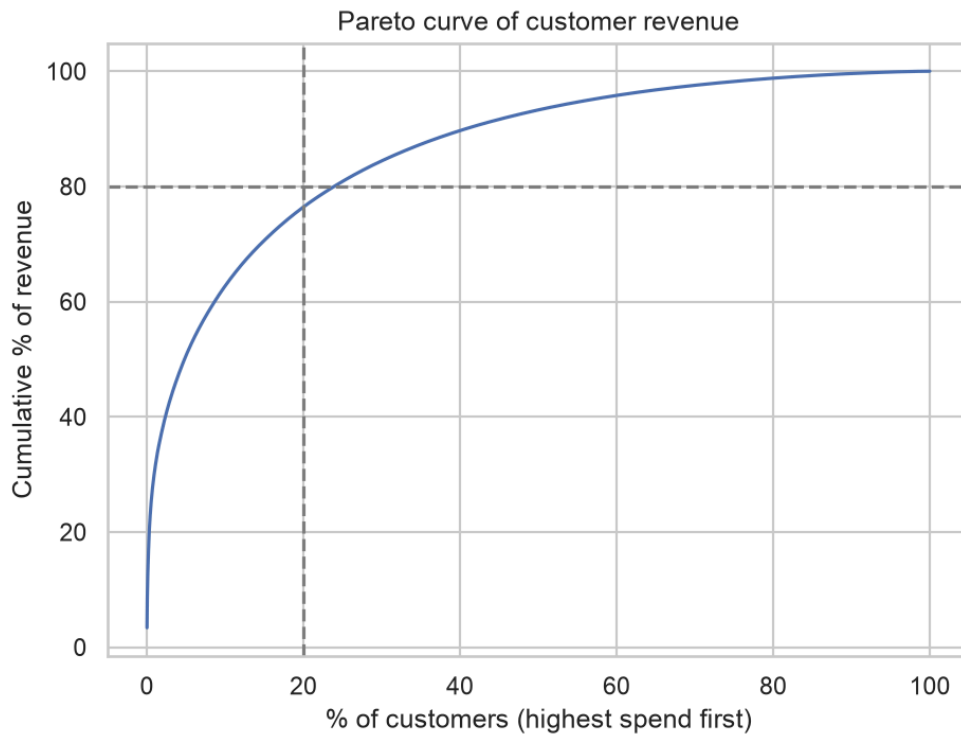


Figure 3.1: Revenue concentration. The top 10% of customers account for 62.7% of revenue.

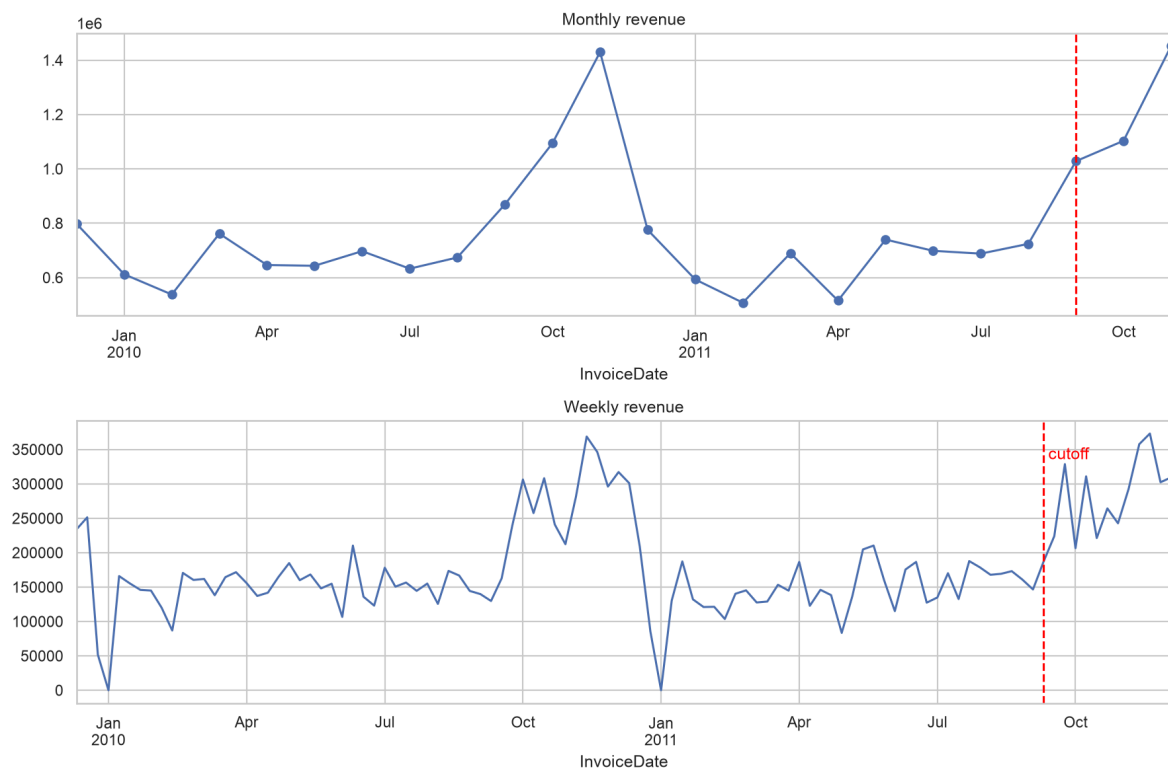


Figure 3.2: Monthly revenue. November peaks are about 2.2 times a typical off-peak month.

3.6 Initial business insights

Finding	What it means for the business
Revenue is concentrated: the top 1% of customers = 30.8% of revenue; top 10% = 62.7%; top 10 customers = 16.1%	Losing a handful of accounts would hurt materially. Retention effort should be weighted towards high-value customers (Task 12, S4)
Most customers are inactive at any moment: only 43.4% repurchased in the 90 days; 30.0% have ordered once; median recency is 162 days	This gap is the retention opportunity the project targets
The “average customer” is misleading: mean spend is 3.3 × the median	Track medians and segments, not averages
UK-dependent, but with valuable international accounts: the UK = 85.3% of revenue, yet 4 of the top 10 customers are international while only 8.7% of customers are	Hypothesis for Task 4: international customers place larger orders
Strong seasonality: November revenue £1.43M (2010) and £1.45M (2011), about 2.2 × the median off-peak month (£660k); the rise starts in August/September	Stock, staff and cash need building from August. The 90-day target window (Sep–Dec 2011) falls inside the peak, so the 43.4% repurchase rate probably overstates the off-peak rate
B2B trading rhythm: Saturday = 0.1% of revenue; 75% of revenue arrives between 10:00 and 16:00	Trade customers order in business hours: time campaigns for weekday mornings
Cancellations are concentrated: mean cancellation rate 11.4%, median 0, 90th percentile 38%	A minority of customers generate most cancellations: a candidate risk signal (used as a model feature)

References (Task 3)

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Task 4 – Statistical Inference

Owner: Member B. All results come from `notebooks/r/02_statistical_inference.R`, run on the customer table ($n = 5,253$). These are observational comparisons: they establish associations, not the effect of any intervention.

Four pre-planned tests answer four different client questions. They are not a search through many comparisons, so no correction is applied across the four headline tests (with Bonferroni the threshold would be 0.0125, and every headline result would still pass). Within Test 4, Games–Howell controls the error rate across its three pairwise comparisons. Every test reports an effect size with a confidence interval, because with 5,253 customers even trivial differences become “significant” (Lin et al., 2013; Sullivan & Feinn, 2012).

Summary

Test	Question	Result	Effect size [95% CI]	Size
1. Means	Do international and UK customers differ in typical order value?	Welch $t(515.8) = 12.41$, $p < .001$	Geometric-mean ratio 1.68 [1.55, 1.82]; $d = 0.72$ [0.62, 0.81]	Medium–large
2. Proportions	Do Q4-acquired customers repurchase more often?	$\chi^2(1) = 43.37$, $p < .001$	+9.6 points [6.7, 12.5]; RR 1.24 [1.16, 1.32]; $V = 0.09$ [0.06, 0.12]	Small
3. Variances	Is order value more variable for international customers?	Brown–Forsythe $F(1, 5251) = 123.94$, $p < .001$	Variance ratio 5.24, bootstrap [1.94, 13.39]	Large but imprecise
4. ANOVA	Does early spending rate differ by acquisition cohort?	Welch $F(2, 1343.7) = 45.11$, $p < .001$	$\omega^2 = 0.022$ [0.014, 0.030]	Small

Test 1 – Comparison of means: order value, international vs UK

- **Research question.** Do international and UK customers differ in their typical average order value?
- **Hypotheses.** H_0 : mean log(average order value) is equal for UK and international customers. H_1 : the means differ (two-sided; no direction was assumed in advance).
- **Why this test.** Average order value is heavily right-skewed (Task 3), so it is compared on the log scale. A difference in mean logs back-transforms to a **ratio of geometric means**, which is easy to read in business terms. The groups are very unequal in size (457 vs 4,796) and Test 3 shows their variances differ, so **Welch’s t-test** is used instead of Student’s (Delacre et al., 2017).
- **Assumption checks.** On the log scale both groups are close to symmetric (skewness 0.29 international, –0.39 UK). With groups this large, the t-test needs only the group means to be approximately normal. A Mann–Whitney U test, which does not assume normality, is run as a robustness check.
- **Results.** Geometric-mean order value: international **£432**, UK **£257**. Welch $t(515.8) = 12.41$, $p < .001$. **Geometric-mean ratio (international / UK) = 1.68, 95% CI [1.55, 1.82]**. Cohen’s $d = 0.72$ [0.62, 0.81]. Mann–Whitney agrees: $W = 1,495,183$, $p < .001$.
- **Interpretation.** A typical international customer’s average order is about 55–82% larger than a typical UK customer’s. This is a medium-to-large difference, and it does not depend on the log transform.
- **Practical implication.** Pricing and account management should not assume overseas orders are smaller. The opposite holds, which is consistent with overseas wholesale accounts consolidating orders to save on freight.
- **Limitation.** Country is observed, not assigned, so this is an association. The international group mixes about 40 countries of very different sizes.

Test 2 – Comparison of proportions: repurchase, Q4-acquired vs others

- **Research question.** Do customers first acquired in October–December repurchase in the 90 days after the cutoff at a different rate from other customers?
- **Hypotheses.** H0: the repurchase proportion is equal for Q4-acquired and other customers. H1: the proportions differ (two-sided).
- **Why this test.** Both variables are binary, so a chi-square test of independence on the 2×2 table (equivalent to the two-proportion z-test) with Yates' continuity correction is appropriate.
- **Assumption checks.** Customers are independent (one row each), and the smallest expected cell count is 749, far above the required 5.
- **Results.** Repurchase rate: **49.8%** for Q4-acquired customers (861 of 1,728) vs **40.2%** for others (1,417 of 3,525). $\chi^2(1) = 43.37$, $p < .001$. **Difference = 9.6 percentage points, 95% CI [6.7, 12.5]**. Relative risk 1.24 [1.16, 1.32]; odds ratio 1.48 [1.32, 1.66]; Cramér's $V = 0.09$ [0.06, 0.12].
- **Interpretation.** The difference is real but **small**. The tiny p-value reflects the large sample more than a strong effect.
- **Practical implication.** A Q4-acquired customer is about 10 points more likely to come back. That is worth using as one input to a targeting model (it is one of the Task 5 predictors), but it is too weak to justify a budget decision on its own.
- **Limitation.** The outcome window (Sep–Dec 2011) is itself the Christmas season. The association may reflect a seasonal buying rhythm (customers first seen in Q4 buy again in Q4) rather than anything about *when* they were acquired, and this design cannot separate the two.

Test 3 – Comparison of variances: spread of order value, international vs UK

- **Research question.** Is customer-level average order value more variable among international customers than among UK customers?
- **Hypotheses.** H0: the variance of average order value is equal in the two groups. H1: the variances differ (two-sided; the direction is read from the variance ratio).
- **Why this test.** The **Brown–Forsythe** (median-centred Levene) test is robust to non-normality (Brown & Forsythe, 1974). The classical F-test of variances is not, and average order value is heavily skewed with very unequal groups.
- **Assumption checks.** Descriptive statistics: international $n = 457$, mean £664, SD £966; UK $n = 4,796$, mean £334, SD £422. Median centring protects the test against the skew and outliers in both groups.
- **Results.** Brown–Forsythe $F(1, 5251) = 123.94$, $p < .001$. **Variance ratio (international / UK) = 5.24, bootstrap 95% CI [1.94, 13.39]** (5,000 resamples). The bootstrap is used because a parametric interval would rest on the normality assumption the test was chosen to avoid.
- **Interpretation.** The interval lies entirely above 1, so international customers' order values are more variable. The interval is **wide**, because there are only 457 international customers and a few very large accounts dominate the variance. The size of the difference (anything from about 2× to 13×) is therefore uncertain, even though its direction is clear.
- **Practical implication.** A single “average international order” figure is unreliable for planning. International accounts should be segmented by order size (Task 12, O4).
- **Limitation.** The test shows a difference in spread, not its cause, and not whether any pricing or credit policy would change it.

Test 4 – ANOVA: early spending rate by acquisition cohort

Research question. Does customers' spending rate over their first 90 days differ across acquisition-year cohorts?

Design history. An earlier version compared total spend (Monetary) by Frequency quartile, which is partly circular because Monetary = Frequency × average order value. Comparing *total* spend by cohort is confounded by exposure time: the 2009 cohort had about 640 days to accumulate spend, the 2011 cohort about 138. Dividing by tenure was checked and rejected: 2 customers have zero tenure, 102 have under 30 days, and $\log(\text{Monetary} / \text{Tenure})$ is *negatively* correlated with tenure ($r = -0.24$), so it reverses the bias rather than removing it.

Outcome. **SpendRate90** = spend in each customer's first 90 days after their first purchase, divided by 90 (£ per day). Every customer is measured over the same 90-day exposure. Only customers whose full 90-day window ends before the cutoff are eligible, which **excludes 302 customers acquired in the 90 days before the cutoff, all from the 2011 cohort**. The outcome is analysed as $\log(\text{SpendRate90})$.

Hypotheses. H0: mean $\log(\text{SpendRate90})$ is equal across the 2009, 2010 and 2011 cohorts. H1: at least one cohort mean differs.

Why this test. Cohort sizes are very unequal and the variances differ (checked below), so **Welch's one-way ANOVA** is used instead of the classical F-test (Delacre et al., 2019). Games–Howell post-hoc comparisons allow for unequal variances and sample sizes. Kruskal–Wallis is run as a robustness check, and ω^2 is reported as the effect size.

Assumption checks.

Cohort	n	Mean rate (£/day)	SD	Median	Mean log	Skew (log)
2009	951	16.40	58.58	6.30	1.870	0.28
2010	3,364	7.56	16.12	4.29	1.492	−0.11
2011	636	7.10	13.91	3.84	1.412	−0.11

On the log scale the outcome is close to symmetric within every cohort. 52 observations lie more than 3 SD from their cohort mean and are kept. Brown–Forsythe: $F(2, 4948) = 23.91$, $p < .001$, so the variances differ, which supports Welch. The residual Q-Q plot is roughly symmetric with heavier tails than normal (Figure 4.1), which is why the rank-based check matters.

Results. **Welch $F(2, 1343.7) = 45.11$, $p < .001$. $\omega^2 = 0.022$, 95% CI [0.014, 0.030]: a small effect.** Cohort accounts for about 2% of the variation in log spending rate.

Comparison (Games–Howell)	Ratio of geometric-mean rates [95% CI]	Adjusted p
2010 vs 2009	0.69 [0.62, 0.76]	< .001
2011 vs 2009	0.63 [0.55, 0.72]	< .001
2011 vs 2010	0.92 [0.83, 1.02]	0.157

Robustness: Kruskal–Wallis $\chi^2(2) = 91.39$, $p < .001$, $\eta^2[H] = 0.018$ (small). The two tests agree.

Sensitivity: what total spend would have shown. On the same customers, $\log(\text{Monetary})$ gives Welch $F(2, 1411.0) = 317.88$ and $\omega^2 = 0.126$ [0.109, 0.143], about six times the effect. Most of the cohort difference in *total* spend is explained by older cohorts having had longer to accumulate it.

Interpretation. The whole effect comes from the 2009 cohort, whose geometric-mean early spending rate is about 1.5 times that of 2010 and 2011. The 2010 and 2011 cohorts do not differ significantly. Because the data begin on 1 December 2009, the “2009 cohort” contains every customer active that month, including established

accounts acquired before the data start. Its higher rate is therefore best read as “established accounts spend faster”, not as “customers acquired in 2009 were better”.

Practical implication. There is no evidence that recently acquired customers spend more slowly in their first 90 days than the 2010 cohort did, so this gives no reason to change acquisition strategy. Cohort alone is a weak basis for segmentation.

Limitations. Tenure normalisation reduces the mechanical advantage of earlier cohorts (more time to accumulate spending) but does not remove all cohort-related confounding. The 2009 cohort is **left-censored** (see above). The 2011 cohort is **right-censored**: only customers acquired January to early June 2011 can be followed for 90 days. The 90-day windows fall in different seasons. ω^2 and its CI come from the classical ANOVA decomposition, which assumes equal variances. The data are observational, so cohort is *associated* with spending rate but does not cause it.

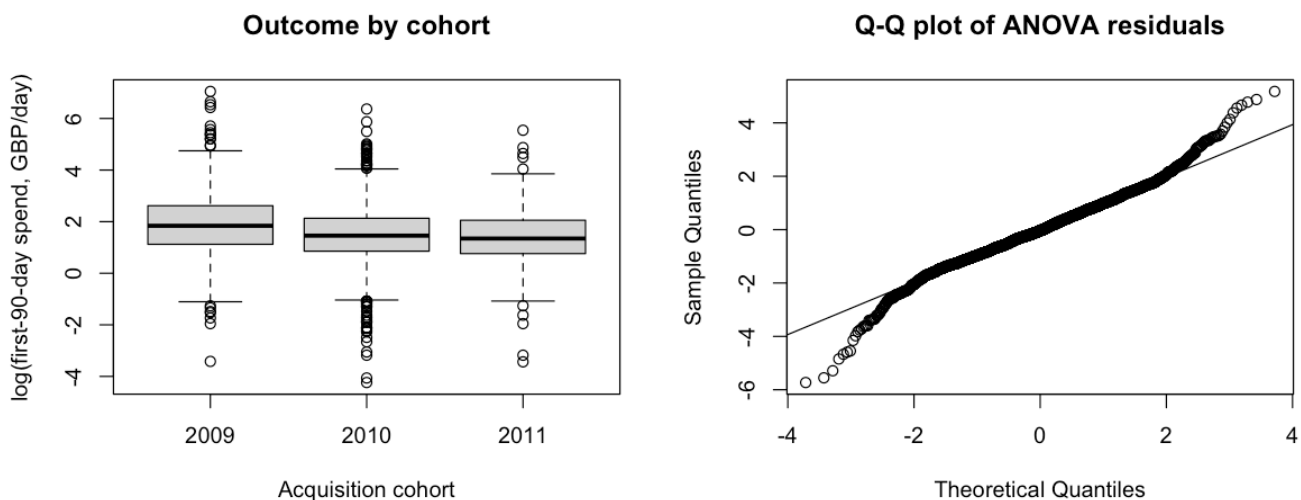


Figure 4.1: Test 4 outcome by cohort (left) and Q-Q plot of the residuals (right).

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Task 5 – Predictive Statistical Modelling

Owner: Member C. The formal model selection comes from `notebooks/python/03b_model_selection.ipynb` (sections D0–D8), and its tables are exported to `reports/tables/task5_*.csv` and `.md`. Fitting details, assumption checks and diagnostics come from the exploratory analysis in `notebooks/python/03_predictive_modelling.ipynb` (sections A1–C4).

5.1 Three related prediction problems

Problem	Target	Evaluated on	Candidates
1. Repurchase probability	Repurchase : any purchase in the next 90 days	All eligible customers	Full logistic; forward, backward and best-subset selection (AIC); Ridge; LASSO; elastic net. Baselines: constant training rate, recency rank
2. Conditional positive spend	FutureSpend among customers who buy	Repurchasers only	OLS on log(spend) with Duan smearing; Gamma GLM (log link); training mean (training median shown for illustration only)
3. Unconditional expected spend	$P(\text{repurchase}) \times E[\text{spend} \text{repurchase}]$	All eligible customers, including £0 spenders	Each spend model combined with the selected probability model; constant training mean as baseline

These are different questions, so **no single “best model” is declared across them**.

Features (9). Recency, TenureDays, CancellationRate, $\log_{10}(\text{Frequency})$, $\log_{10}(\text{Monetary})$, $\log_{10}(\text{DistinctProducts})$, $\log_{10}(\text{AvgBasketValue})$, IsUK and AcquiredInQ4. All are computed from purchases **before** the origin t . Both targets use the **half-open window $[t, t + 90 \text{ days})$** . CustomerID and the targets are never predictors.

The probabilities are **predictions under the conditions observed historically**, including whatever marketing the retailer was already doing. They are **not** identified “no-offer” probabilities. Historical data identify neither the effect of a campaign nor what customers would have done without one. That requires the Task 6 experiment.

5.2 Validation design: why not just a random split

A random customer split, used in the exploratory notebook 03, tests whether the model predicts *other customers in the same period*. The retailer will instead train on the past and score a *later* period. Model selection therefore uses **rolling-origin development periods**. Origins are spaced exactly 90 days apart, so each training snapshot’s labels are complete by the next origin, when the model would be used:

Role	Train on origin	Predict at origin	Eligible customers	Repurchase prevalence
Development period 1	16 Jun 2010	14 Sep 2010	3,364	57.5%
Development period 2	14 Sep 2010	13 Dec 2010	4,292	30.4%
Development period 3	13 Dec 2010	13 Mar 2011	4,581	35.3%
Development period 4	13 Mar 2011	11 Jun 2011	4,951	32.0%
Retrospective temporal evaluation	11 Jun 2011	9 Sep 2011	5,253	43.4%

- **Development only for selection.** All tuning, subset selection, scaling and smearing happen inside each training snapshot. For Ridge, LASSO and elastic net the scaler sits *inside* a `GridSearchCV` pipeline: the regularisation strength is chosen by 5-fold inner cross-validation on the **Brier score**, so no inner validation fold influences the scaling. Forward, backward and best-subset selection are re-run by AIC in every training snapshot.
- **The four development periods cover each season once**, which is why they are weighted equally.

- **Prior exposure, stated honestly.** The Sep–Dec 2011 outcomes were examined in earlier versions of this analysis, so they cannot be an untouched test set. The final evaluation is a **retrospective temporal evaluation** of a model locked on development data. The dataset contains no later, unexamined data.
- **Repeated customers.** The same customer appears in several snapshots with different, time-stamped features. For a model meant to score *returning* customers, that is intended and is not leakage. But the rows are not independent, so uncertainty that pools periods uses a bootstrap that resamples **customers**.

5.3 Metrics and the selection rule (declared before the results)

Metric	Role	Better
Brier score	Primary: the probabilities feed the two-part expected-spend calculation	Lower
Log-loss	Secondary; penalises confident errors	Lower
ROC-AUC	Ranking across all thresholds	Higher
Average precision (AP)	Precision–recall summary; depends on prevalence	Higher
Calibration intercept / slope	Logistic regression of the outcome on the predicted log-odds, estimated jointly	≈ 0 / ≈ 1
Precision@10%, Lift@10%	Repurchase rate among the top $\text{ceil}(0.10 n)$ scores (ties broken by CustomerID); lift = precision $\div$ prevalence. 10% is an illustrative contact capacity, not a client budget	Higher
Predictors	Non-zero coefficients, excluding the intercept (	β

Brier and log-loss measure overall probabilistic accuracy, not calibration alone. Accuracy, F1, a 0.5 threshold and scenario profit are not used for selection: there is no fixed classification decision, and profit depends on an unmeasured uplift.

Rule.

1. Exclude any candidate with a non-converged fit or invalid predictions.
2. Select the candidate with the smallest unrounded mean development Brier (equal weight per period).
3. Break exact ties by fewer predictors, then lower log-loss, then a fixed candidate order.
4. Report clustered bootstrap uncertainty.
5. Lock the selected candidate, refit it on the final training snapshot, and report its evaluation without reselecting.

All seven candidates converged in every period; none was excluded.

5.4 Repurchase probability: development results (used for selection)

Model	Predictors (mean)	Brier ↓	ROC-AUC ↑	AP ↑	Cal. intercept / slope	Precision@10% ↑	Lift@10% ↑	Verdict (Brier difference vs LASSO, 95% CI)
LASSO	7.2	0.2026	0.779	0.713	−0.30 / 0.96	0.864	2.35	Selected under the Brier rule
Ridge	9	0.2027	0.778	0.713	−0.31 / 0.96	0.864	2.35	Indistinguishable: +0.00001 [−0.00014, +0.00016]
Elastic net	7.5	0.2028	0.778	0.713	−0.30 / 0.97	0.863	2.34	Worse: +0.00014 [+0.00005, +0.00023]

Model	Predictors (mean)	Brier ↓	ROC-AUC ↑	AP ↑	Cal. intercept / slope	Precision@10% ↑	Lift@10% ↑	Verdict (Brier difference vs LASSO, 95% CI)
Forward selection	5.5	0.2030	0.779	0.713	−0.30 / 0.95	0.862	2.34	Worse: +0.00035 [+0.00009, +0.00061]
Backward selection / best subset	5.5	0.2030	0.779	0.713	−0.30 / 0.95	0.862	2.34	Worse: +0.00037 [+0.00011, +0.00064]
Full logistic	9	0.2032	0.778	0.712	−0.31 / 0.92	0.864	2.35	Worse: +0.00051 [+0.00024, +0.00079]
Constant (training rate)	—	0.2471	0.500	0.388	N/A	0.414	1.08	Baseline
Recency rank	—	N/A	0.719	0.587	N/A	0.670	1.81	Baseline

Development: equal-weight means over four rolling-origin periods. CIs come from 2,000 bootstrap replicates resampling 4,951 distinct customers across all periods.

Reading the table. Every model is far better than the baselines. The differences *between* models are tiny: at most 0.0005 in Brier (about 0.25%), with identical AUC to two decimals. LASSO is the model **selected under the declared rule**, not a clearly superior model, and Ridge is statistically indistinguishable from it. LASSO kept 7.2 of 9 predictors on average. Recency, cancellation rate and log(Frequency) were selected in every fit with stable signs. Tenure, log(Monetary), log(DistinctProducts) and Q4 acquisition changed sign across fits, because they are correlated (see 5.6). Their individual coefficients should not be over-interpreted.

5.5 Retrospective temporal evaluation of the locked model

LASSO was refitted on the 11 Jun 2011 snapshot ($C = 0.089$, 5 predictors kept) and evaluated on the 9 Sep 2011 customers.

Model	Role	Brier	Log-loss	ROC-AUC	AP	Cal. intercept / slope	Precision@10%	Lift@10%
LASSO	Locked	0.2005	0.608	0.791	0.758	+0.63 / 0.79	0.899	2.07
Full logistic	Benchmark	0.2014	0.612	0.789	0.756	+0.60 / 0.76	0.895	2.06
Constant (training rate)	Benchmark	0.2584	0.712	0.500	0.434	N/A	0.462	1.07
Recency rank	Benchmark	N/A	N/A	0.762	0.681	N/A	0.768	1.77

Prevalence 0.434; $n = 5,253$. Other candidates are reported in `reports/tables/task5_final_classification.md` for description only. Nothing is reselected from them.

Paired bootstrap (2,000 replicates), locked LASSO minus benchmark:

- vs full logistic: Brier −0.0009 [−0.0012, −0.0005]; AUC +0.002 [+0.001, +0.003].
- vs constant rate: Brier −0.058 [−0.064, −0.052].
- vs recency rank: AUC +0.030 [+0.023, +0.037]; Precision@10% +0.131 [+0.095, +0.169].

Ranking holds up in the later period, but calibration does not. The locked model predicts a mean of 0.30 against an actual 0.43 (intercept +0.63, slope 0.79). Across the development periods the calibration intercept ranged from −1.76 to +0.40: the probability *level* follows the training season. One recalibration policy was tested on development data only: an intercept shift learned from the most recent completed period. It made the evaluation **worse** (Brier 0.2005 → 0.2063), because the seasonal shift reversed. **No simple recalibration policy is supported by these data.**

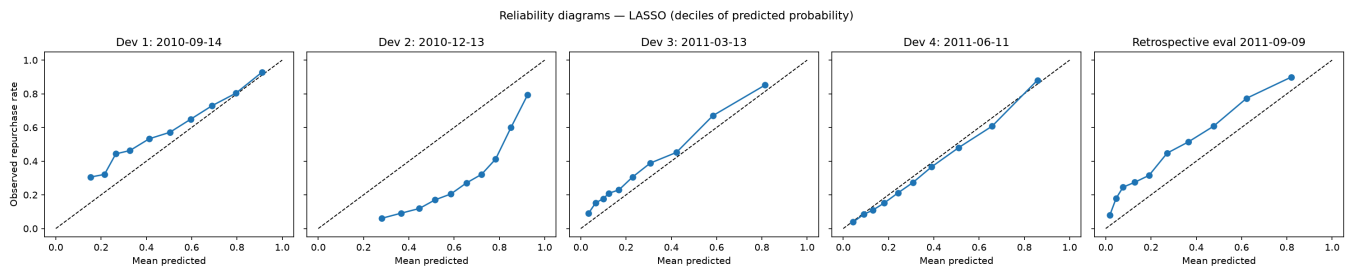


Figure 5.1: Reliability diagrams for the selected model in each development period and the retrospective evaluation.

5.6 Assumptions and diagnostics (exploratory notebook 03)

- **Multicollinearity.** VIF: $\log_{10}(\text{Monetary})$ 203, $\log_{10}(\text{Frequency})$ 98, $\log_{10}(\text{AvgBasketValue})$ 59 (all others < 3.1). $\text{Monetary} = \text{Frequency} \times \text{AvgBasketValue}$, so the identity holds exactly on the log scale. For the **log1p** features used here it is not exact, but very close: a regression of $\log_{10}(\text{Monetary})$ on the other two has $R^2 = 0.994$. In the unpenalised model this makes the coefficients of $\log_{10}(\text{Frequency})$ and $\log_{10}(\text{Monetary})$ unstable (odds ratios 3.48 [1.43, 8.48] and 0.51 [0.15, 1.74]).
- **Ridge vs LASSO.** Ridge keeps all nine predictors and stabilises the correlated coefficients through **shrinkage**, which is what it was designed for (Hoerl & Kennard, 1970). LASSO additionally sets some coefficients to zero (Tibshirani, 1996). That sparsity is a separate interpretability advantage, not automatic superiority, and the two are statistically indistinguishable here (5.4).
- **Linearity of the logit.** Close to linear for frequency, monetary value and product breadth. Recency is mildly curved (Box–Tidwell $p = 0.006$), and tenure is U-shaped.
- **Influence.** No customer dominates the repurchase model (largest Cook's distance 0.008).
- **Class balance.** At 43/57 (and 30–58% across periods), no resampling is needed. Resampling would also distort the probabilities the expected-spend calculation relies on.

5.7 Spend: conditional and combined selection

Conditional positive spend (repurchasers; primary metric RMSE in £, because the quantity needed is a conditional mean):

Model	Development RMSE	MAE	Median AE	Retrospective eval. RMSE (n = 2,278)	MAE	Median AE	Mean predicted vs observed
Log-OLS + Duan smearing	2,062	636	296	3,358	657	222	£808 vs £1,223
Gamma GLM (log link)	2,074	638	298	3,398	668	218	£786 vs £1,223
Training mean	3,123	999	704	4,694	1,090	673	£1,042 vs £1,223
Training median (illustrative, targets the median)	3,169	796	279	4,756	955	274	£431 vs £1,223

Unconditional expected spend (all customers, with the LASSO probability fixed):

System	Development RMSE	MAE	Retrospective eval. RMSE (n = 5,253)	MAE	Median AE	Mean predicted vs observed
P × Log-OLS + Duan smearing	1,377	405	2,269	400	101	£294 vs £530
P × Gamma GLM	1,386	404	2,297	406	104	£287 vs £530
P × training mean	1,991	547	3,098	504	187	£314 vs £530

System	Development RMSE	MAE	Retrospective eval. RMSE (n = 5,253)	MAE	Median AE	Mean predicted vs observed
Constant training mean (baseline)	2,039	611	3,154	613	334	£334 vs £530

- **Log-OLS with Duan smearing is selected for both problems.** The Gamma GLM is the more natural model *in principle* for a positive outcome whose variance rises with the mean (Manning & Mullahy, 2001), and it is close behind, but the declared rule picks the lower RMSE.
- **The selected systems clearly beat their training-mean baselines in the later period.** Conditional RMSE is lower by £1,336 (95% CI £693–£1,952) and expected-spend RMSE by £885 (£464–£1,293). Both **under-predict the level** in the Q4 evaluation window, so the £ amounts are not calibrated across seasons.
- **Large spenders drive RMSE.** The top 1% of evaluation repurchasers account for 28.4% of their spend. One training customer (1 order, £14 of history, then £450 of spend) has Cook's distance 7.4 in the Gamma GLM. Removing the five most influential *training* customers lowers Gamma's evaluation RMSE from £3,398 to £3,279. All customers are kept in the primary comparison.

5.8 From prediction to decisions

A retention campaign creates value only through the **incremental** purchases it causes (Ascarza, 2018; Devriendt et al., 2021). These models do not estimate that. The exploratory notebook therefore keeps three analyses apart:

1. **Value ranking:** who has the highest expected spend. This needs no assumption about offers.
2. **Retrospective validation (notebook 03, A5 and C2):** each ranking scored on the test customers' realised spend. That spend only exists after the window closes, so this validates the ranking. It is neither a deployable profit nor an upper bound, because it still rests on an assumed uplift. Contacting the top 10% by repurchase probability would have earned about £9.3k, and the top 10% by expected spend about £9.8k, against £6.5k for contacting everyone (assumptions: 30% margin, £10 offer, 10% uplift).
3. **Prospective decision rule (notebook 03, C3):** contact a customer if **uplift × margin × expected spend > offer cost**. The threshold comes from the economics, not from a 0.5 probability cut-off: with the placeholder values, expected 90-day spend must exceed £333. That selects 39.8% of test customers, for a model-predicted incremental profit of £13.2k, **conditional on the assumed uplift**. Across uplift × margin from 1% to 8% and offer costs of £5–£20, the contact share runs from **4% to 82%** and predicted profit from **£1.4k to £47k**. Because the spend *level* is under-predicted in a new season (5.7), these figures are scenarios, not forecasts.

5.9 Recommendation

- **Probability model: LASSO logistic regression**, selected under the pre-declared Brier rule and locked before the retrospective temporal evaluation. Ridge is an equally defensible alternative (statistically indistinguishable). The choice does not rest on AUC: all candidates tie to two decimals.
- **Use it to rank customers.** Ranking is supported in the development periods and in the later period. It beats the recency rule (AUC +0.030; precision in the top 10% +13 points) and the constant rate.
- **Do not treat its probabilities, or the spend predictions, as calibrated £ quantities across seasons.** Their level shifted in every period, and a simple recalibration made things worse. Any budget calculation needs a same-season or experimentally validated correction.
- **Spend: log-OLS with Duan smearing**, for both conditional and expected spend.

Limitations.

- The models predict repurchase, not response to an offer.

- The evaluation period was seen before, so the evaluation is retrospective, not blind.
- There are only four development periods and one evaluation period, from about two years of data.
- Monetary still counts orders that were later cancelled.
- Conclusions apply only to identifiable customers (Task 3).

References (Task 5)

Full entries are in the Task 2 references (section 2.9): Ascarza (2018); Devriendt et al. (2021); Hoerl & Kennard (1970); Manning & Mullahy (2001); Tibshirani (1996).

Task 6 – Critical Evaluation of Experimental Design

Owner: Member B. Numbers come from `notebooks/python/07_experimental_design.ipynb` (sample size, power and blocking) and from notebooks 03 and 05 where stated. Re-check them if notebook 00 or `src/` changes.

6.1 Why an experiment is needed

Every model in this project predicts **who will repurchase under the conditions seen historically**. None of them measures **what a retention offer changes**. The difference matters commercially:

- The repurchase models (Task 5) rank customers well (repeated cross-validation AUC 0.799 ± 0.015 ; top-decile lift 2.09), but the customers they rank highest are mostly those who were going to buy anyway. The top recency-by-frequency tier repurchases at 81% with no offer at all (notebook 07).
- The profit calculations in notebooks 03 and 05 depend on an **assumed** uplift. Across plausible values of uplift $\times$ margin, the recommended contact share ranges from 4% to 82%, and predicted profit from £1.4k to £47k (notebook 03, C3). The data cannot narrow this range. Customers were never randomly offered anything, so the offer's effect is not identified.
- The literature makes the same point. The highest-risk customers are not necessarily the ones an intervention helps (Ascarza, 2018), and retention profit comes from the incremental effect of the offer, not from the churn probability (Lemmens & Gupta, 2020; Devriendt et al., 2021). Randomised controlled experiments are the standard way to establish that effect (Kohavi et al., 2009).

An experiment is therefore not optional polish. It is the only way to turn the Task 5 ranking into a defensible budget. The consultancy is not required to run one (brief, Task 6). This section evaluates which design the retailer should use.

6.2 The two designs

Completely Randomised Design (CRD). Every eligible customer is independently assigned to treatment or control with the same probability, ignoring any customer characteristic (Montgomery, 2017).

Randomised Complete Block Design (RCBD). Customers are first grouped into blocks that are expected to differ in the outcome. Every treatment appears within every block, and randomisation happens separately inside each block. Differences between blocks are then removed from the error term (Montgomery, 2017).

Both designs share the same core set-up:

Element	Specification (both designs)
Experimental unit	The customer account (<code>CustomerID</code>). This is the level at which offers are sent and at which repurchase is recorded.
Treatment arms	(T) a retention offer, e.g. a £10 voucher on the next order, sent by e-mail. An optional third arm, “contact without discount”, would separate the effect of <i>being contacted</i> from the effect of <i>the discount</i> , but it costs sample size (see 6.4).
Control group	No contact, business as usual. A concurrent control is essential. The repurchase rate varies strongly by season (32% in the summer 2011 window vs 43% in the Sep–Dec 2011 window, notebook 03 A8), so a before/after comparison would confuse the offer with the season.
Primary outcome	Repurchase (yes/no) within 90 days of the send date. This is the same definition as the modelling target, so results feed straight back into the models.
Secondary outcomes	Spend in the 90 days (reported with a confidence interval, not tested as primary; see 6.4); voucher redemption; unsubscribes and complaints.

Element	Specification (both designs)
Analysis	Difference in repurchase proportions with 95% CI (as in Task 4, Test 2). For the RCBD, a logistic regression with treatment and block effects. The primary outcome, the success threshold and the analysis are fixed before the pilot starts.

Comparison

	CRD	RCBD
Assignment mechanism	Simple random assignment of each customer (e.g. a seeded random number per CustomerID), with fixed arm proportions	Random assignment separately within each block, with the same arm proportions in every block
Suitable blocking variables	None used	Recency tier × frequency tier (9 blocks), which are known before the pilot and strongly related to the outcome (repurchase ranges from 14% to 81% across blocks). Alternatives: the Task 5 model-score tier; UK vs international; acquisition quarter (Task 4, Test 2 shows Q4-acquired customers repurchase 9.6 points more)
Advantages	Simplest to implement, explain and audit. No modelling needed at assignment. Valid inference by construction	Removes between-block variation, so the same power needs about 23% fewer customers (notebook 07). Guarantees balance on the blocking variables in a small pilot. Gives a treatment effect per block , which is exactly what targeting needs (does the offer work better for lapsing customers than for loyal ones?)
Limitations	Wastes power: the huge variation between loyal and lapsing customers stays in the error term. A small pilot can end up unbalanced by chance	Blocks must be defined before assignment and cannot be changed afterwards. With many blocks, some become small (the old-recency/high-frequency block has only 122 customers). Per-block effects need far more data than the overall effect
Sample size / power (5-point uplift, 5% two-sided, 80% power)	1,558 customers per arm	about 1,193 per arm
Seasonality	Controlled by the concurrent control group, but a pilot run in Q4 measures a Q4 effect only	Same. Calendar time can itself be a blocking factor if sends are staggered over several weeks
Contamination / interference	Both designs assume one customer's treatment does not affect another's outcome. In a B2B wholesale base that can fail: several buyers at the same shop may hold separate accounts, vouchers can be forwarded, and account managers may treat control accounts differently. Mitigations: single-use codes tied to the account; assignment at the business level where accounts can be linked; keeping sales staff blind to assignment	Same as CRD
Operational feasibility	High. Needs only a random list and the existing e-mail system	High to moderate. Needs the recency and frequency tiers computed from the order history on the send date, which the Task 10 pipeline would already produce
Ethical considerations	See 6.5	See 6.5

6.3 Which outcome can a pilot realistically detect?

This is the most important practical constraint, and it comes straight from the data (notebook 07).

Spend cannot be the primary outcome. 90-day spend has a mean of £530, a standard deviation of £3,148 and a median of £0 (coefficient of variation 5.9). Detecting a 10% rise in mean spend would need about **55,000 customers per arm**, roughly ten times the entire customer base. Detecting a 20% rise needs about 13,800 per arm. Blocking does not help here, because the blocks explain only 3.9% of spend variance: it is dominated by a few very large wholesale buyers.

Repurchase can be the primary outcome:

Uplift in repurchase rate	CRD, customers per arm	RCBD, customers per arm (approx.)
+3 points	4,314	3,303
+5 points	1,558	1,193
+8 points	611	468
+10 points	391	300

What pilots the retailer could actually run can detect (CRD, 80% power):

Pilot	Customers in smaller arm	Minimum detectable uplift
Whole active base, 50/50	2,626	3.9 points
Whole active base, 80% treated / 20% control	1,050	6.1 points
Three highest-value blocks only, 50/50 (baseline 69%)	1,040	5.6 points

Is that small enough to matter? Under the placeholder assumptions (30% margin, £10 offer), an extra repurchase is worth the offer only if the uplift exceeds about **2.7 points** using the mean spend of a repurchaser (£1,223). Using the median spend (£529) the break-even is about **6.3 points**. Customers nudged back by a voucher are more likely to be small buyers than large ones, so the realistic break-even lies between the two. A whole-base pilot can therefore detect effects large enough to be clearly profitable, but not effects near the lower break-even. That is an honest limit of what the data can support, and it is why the margin and offer cost need to be confirmed with the expert (Task 11).

6.4 Implementation challenges

- **Opportunity cost of the control group.** Withholding offers from 20–50% of customers for 90 days means forgoing any real effect on them. An 80/20 split reduces that cost but raises the detectable effect from 3.9 to 6.1 points.
- **A third arm is expensive.** Adding “contact without discount” splits the sample three ways (about 1,750 per arm on the whole base). That raises the detectable effect for each comparison from 3.9 to 4.8 points, before any correction for making two comparisons. It should be added only if separating contact from discount is a priority for management.
- **Timing.** The 90-day outcome window plus about two weeks of set-up means roughly four months from decision to result. A pilot started in Q4 would measure the effect of an offer during the Christmas peak, when customers buy anyway. Run the first pilot outside October–December (e.g. launch in March, read out in June). If it succeeds, repeat in Q4 before rolling out for the peak season.
- **Noncompliance.** Some treated customers will not open the e-mail. The primary analysis should compare arms as *assigned* (intention-to-treat), which measures the effect of the policy of sending the offer.
- **Novelty and repeat exposure.** A one-off pilot measures a first-offer effect. Repeated offers can train customers to wait for discounts, which a single 90-day pilot cannot detect.

6.5 Ethical considerations

- **Fairness of withholding.** The treatment is a discount on optional purchases, not an essential service. Withholding it from a randomly chosen control group for 90 days is low-harm, but it should be time-limited and not repeated on the same customers indefinitely.
- **Targeting fairness.** Offers targeted by predicted value give better prices to some accounts than others. That is common commercial practice, but it should be a deliberate, documented policy, not an accidental by-product of a model. Small customers systematically receive fewer offers.
- **Data protection and marketing consent.** The pilot uses existing purchase history and sends marketing e-mails. The retailer's data protection lead should confirm the lawful basis for using purchase history to target offers, and the marketing-consent rules for e-mailing each type of account, before launch.
- **Transparency.** Voucher terms should be the same for everyone who receives them, and the experiment should not mislead customers (e.g. no false scarcity).

6.6 Recommendation: a pilot RCBD

Item	Recommendation
Design	RCBD, blocks = recency tier × frequency tier (9 blocks) computed from order history on the send date
Arms	Two: £10 offer vs no contact. Add “contact without discount” only if management specifically needs it
Allocation	50/50 within each block, if the retailer can accept withholding the offer from half the base for one quarter; otherwise 80/20, accepting a larger detectable effect (about 6 points)
Sample	The whole active base (about 5,250 accounts at current size)
Primary outcome	Repurchase within 90 days, analysed as assigned; difference in proportions with 95% CI
Timing	Launch outside Q4 (e.g. March), read out after 90 days. Repeat before Q4 if the pilot succeeds
Pre-registered success threshold	Uplift ≥ 4 points with the 95% CI excluding 0, and incremental margin above the offer cost using the confirmed margin (Task 11)
Stop/go	<i>Go</i> : threshold met → roll out targeted offers using the per-block effects. <i>Refine</i> : effect positive but below threshold → test a cheaper or better-targeted offer. <i>Stop</i> : CI includes 0 → do not fund blanket retention offers

What the pilot cannot prove. It measures the average effect of this offer, at this time of year, on this customer base. It does not validate the predictive models themselves, and a null result means *this* offer did not work, not that retention activity cannot. The predictive models (Task 5) remain useful for deciding **whom to include** in the pilot and how to block it. They do not show that offers **cause** repurchase.

References (Task 6)

Ascarza, E. (2018). Retention futility: Targeting high-risk customers might be ineffective.

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Lemmens, A., & Gupta, S. (2020). Managing churn to maximize profits.

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(5), 956–973. <https://doi.org/10.1287/mksc.2020.1229>

Montgomery, D. C. (2017). *Design and analysis of experiments* (9th ed.). Wiley. *[New source, not in the Task 2 list; a textbook, not counted towards the peer-reviewed minimum.]*

Task 7 – Critical Evaluation of Principal Component Analysis

Owner: Member B. Results come from `notebooks/python/04_pca_evaluation.ipynb`.

What was actually run. Two different methods are used, and they are named precisely:

- **Classical PCA** (`sklearn.decomposition.PCA`) on the 9-feature customer table. Each feature is **centred and scaled** (standardised) first, because the features are on very different scales (days, counts, £, rates).
- **Truncated SVD** (`TruncatedSVD`) on the $5,253 \times 4,445$ customer $\times$ product matrix. This is a **PCA-related** method, not PCA. It factorises the matrix **without centring** it. Centring would turn almost every zero in a 98.3%-sparse matrix into a non-zero value, making the matrix dense. The cost of not centring is that the components are not the directions of maximum variance around the mean, so the “variance retained” figure is not directly comparable to a PCA scree plot. The two methods give the same components only when every column already has mean zero.

7.1 Does the customer feature table need dimensionality reduction?

No. PCA needs **6 of the 9 components to retain 90% of the variance**, so there is almost no compression (Figure 7.1). The real problem in this table is the collinearity of $\log(\text{Monetary})$, $\log(\text{Frequency})$ and $\log(\text{AvgBasketValue})$ (VIF up to 203, Task 5), and LASSO and elastic net already resolve it by **dropping** the redundant feature. That keeps every remaining coefficient as a named, business-readable odds ratio.

Why PCA would harm interpretability here. Each principal component is a weighted mix of all nine features. A statement like “a 1 SD increase in PC2 raises the odds of repurchase by 30%” cannot be acted on by an account manager, whereas “customers who have not ordered for longer are less likely to return” can. In regression, components with little variance can still carry the most predictive signal, so dropping them to save dimensions can throw information away (Jolliffe, 1982).

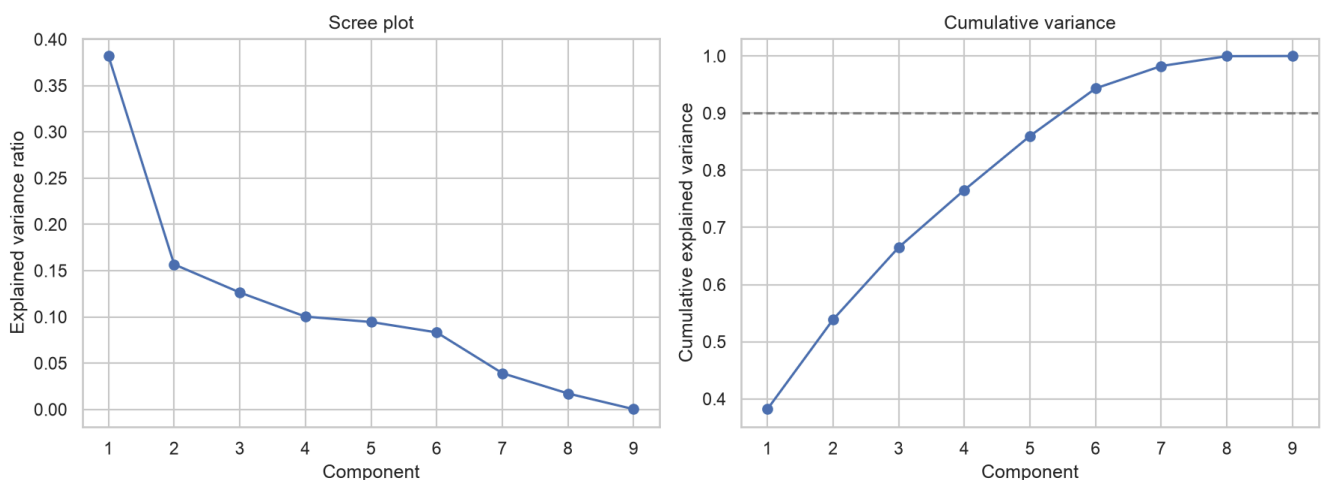


Figure 7.1: PCA on the standardised customer table. Six of nine components are needed for 90% of the variance.

7.2 Where reduction is useful: the sparse customer $\times$ product matrix

Here the dimensionality

is

the problem: 4,445 products, and 98.3% of customer–product cells empty. This is the situation in which dimensionality reduction pays off (Jolliffe & Cadima, 2016).

The input matters as much as the method. The first version ran Truncated SVD on summed **quantities**. Twenty components appeared to retain 86.7% of the variance, but the first component was **100% one customer** (a Danish wholesaler's bulk orders) and the second was 92% another. The components described individual large accounts, not shared buying patterns, because SVD without centring or scaling is dominated by the largest values.

The analysis therefore uses a **binary bought / not-bought matrix**. That asks which products are bought by the same customers, and stops order size from dominating:

	Raw quantities	Binary (used)
Share of SVD1 from the single largest customer	100%	2.7%
Share of SVD2 from the single largest customer	91.6%	0.8%
Variance retained by 20 components	86.7% (misleading)	19.4%

On the binary matrix the components are interpretable:

- **SVD1:** breadth of buying. Customer scores correlate 0.97 with the number of distinct products bought, which is largely the existing DistinctProducts feature.
- **SVD2:** home décor (wicker hearts, wooden frames) vs children's items (lunch bags and boxes, plasters in tins).
- **SVD3:** jumbo and lunch bags vs traditional games and crafts.
- **SVD4:** kitchen and tea-time (Regency teacups, cake stands) vs small novelty items.

The honest compression figure is that 20 components retain only **19.4%** of the variance. Buying across 4,445 products does not reduce to a few dimensions. The components summarise the strongest themes, not the whole picture.

7.3 Impact on predictive modelling

How it was evaluated. On the same train/test split, the same unpenalised logistic model was fitted with and without the 20 SVD components. The SVD was fitted on training customers only, to avoid leakage. The two AUCs were compared with a paired bootstrap (2,000 resamples of the test customers).

Result. AUC rises from **0.8091 to 0.8147**: a difference of **+0.0057, 95% CI [+0.0002, +0.0111]**. The gain is only just distinguishable from zero and small in practice. It costs 20 extra, hard-to-explain features, and it has been checked on one split only, not with cross-validation or out of time as the main model was (Task 5). The product themes are more valuable as **descriptive segmentation**, e.g. for product-specific retention offers, than as a predictive upgrade.

7.4 When PCA should and should not be used

Use PCA / SVD when...	Avoid it when...
Dimensionality itself is the problem (thousands of sparse columns, as with products)	There are only a few features, each already meaningful (the 9-feature table)
The goal is to discover structure, e.g. co-purchase themes for segmentation	Coefficients must be explained to managers as odds ratios
Predictors are numerous and highly correlated and interpretation does not matter	Collinearity can be handled by penalised regression without losing feature identity
Inputs have been prepared so that no single extreme case dominates (binary, scaled or log-transformed)	Raw, heavy-tailed inputs would let a few outliers define the components, as happened with raw quantities here

Conclusion. Dimensionality reduction is not beneficial for this project's customer-level model, but it is useful for describing the product dimension, provided the matrix is prepared carefully. The most transferable lesson is how sensitive it is to preprocessing: on raw quantities, the same algorithm produced components that looked meaningful but were driven by single customers.

References (Task 7)

Jolliffe, I. T. (1982). A note on the use of principal components in regression.

Journal of the Royal Statistical Society: Series C (Applied Statistics), 31

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Task 8 – Critical Evaluation of Bayesian Statistical Methods

Owner: Member C. Results come from `notebooks/python/05_bayesian_methods.ipynb`. The models use the same train/test split and features as Task 5, so the results are directly comparable.

8.1 Naive Bayes

Naive Bayes assumes the features are independent given the outcome. Here that assumption is clearly violated: $\log(\text{Monetary})$, $\log(\text{Frequency})$ and $\log(\text{AvgBasketValue})$ are almost an exact linear combination of one another (VIF up to 203, Task 5).

Test set	AUC	Brier
Naive Bayes (Gaussian)	0.801	0.212
Logistic regression, same split (reference)	0.810	0.175
Constant base-rate prediction	—	0.246

Naive Bayes ranks customers almost as well as logistic regression. It can classify well even when its independence assumption fails (Domingos & Pazzani, 1997). But its **probabilities are badly calibrated**: by treating three correlated features as independent evidence, it counts the same signal two or three times and pushes probabilities towards 0 and 1 (Niculescu-Mizil & Caruana, 2005; Figure 8.1). **Verdict**: acceptable as a quick ranking baseline, but its probabilities must not feed a profit rule without recalibration.

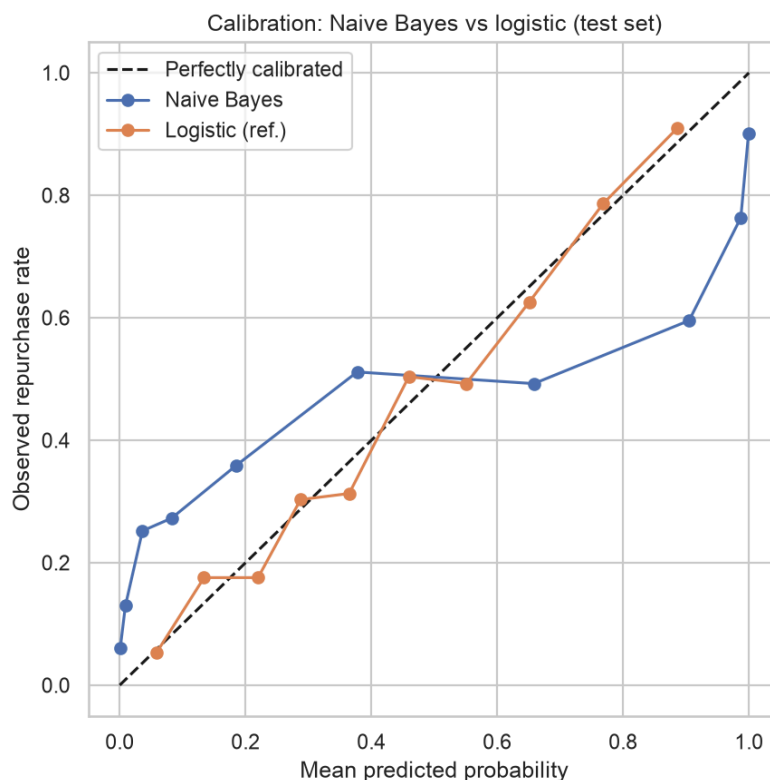


Figure 8.1: Calibration of Naive Bayes vs logistic regression (test set).

8.2 Bayesian hierarchical regression: partial pooling across countries

Why it is needed. The customer base spans **41 countries**; **26 have fewer than 10 customers** and **13 have only 1 or 2**. A separate repurchase rate per country would be meaningless for most of them: one customer

makes the rate 0% or 100%. Ignoring country throws away real differences. Partial pooling sits between the two. Each country gets its own baseline, drawn from a shared distribution, so small countries are pulled (“shrunk”) towards the overall rate while large countries keep their own (Efron & Morris, 1975; Abe, 2009).

Model. $\text{logit } P(\text{repurchase}) = a[\text{country}] + b_{\text{recency}} \times \text{Recency} + b_{\text{freq}} \times \log(\text{Frequency})$ (standardised), with $a[\text{country}] = \mu_a + \sigma_a \times \text{offset}[\text{country}]$. This is a non-centred parameterisation, which samples better.

Priors, and why.

- $\mu_a \sim \text{Normal}(0, 1.5)$: the overall baseline log-odds. It puts most prior mass between about 5% and 95% repurchase, so nothing plausible is ruled out.
- $\sigma_a \sim \text{HalfNormal}(1)$: how much countries differ. An SD of 1 on the log-odds scale already allows rates from roughly 20% to 70%, so this says “differences of that size are plausible, much larger ones are unlikely”. A weakly informative prior on a between-group SD is the recommended choice when there are few groups or little data per group (Gelman, 2006).
- $b \sim \text{Normal}(0, 1)$: weakly informative. With about 3,900 customers the data dominate: the posterior SD is about 0.05.

Convergence. The first run (`target_accept 0.9`) produced **26 divergences**. These came from the funnel-shaped posterior near $\sigma_a = 0$, which is plausible here because the data cannot rule out countries barely differing. The final run uses `target_accept 0.99` with 2,000 tuning and 2,000 sampling iterations in each of 4 chains: **0 divergences out of 8,000 draws, maximum R-hat 1.004, and bulk ESS $\geq 1,309$** (for σ_a , the hardest parameter). The posterior means barely changed between the two runs.

Posterior results and uncertainty.

- Recency: -0.75 ; $\log(\text{Frequency})$: $+0.70$ (posterior means on the log-odds scale). The signs agree with Task 5.
- **σ_a : mean 0.46, 89% interval [0.06, 0.99].** The data cannot tell whether countries barely differ or differ substantially. That is an honest reflection of how little information most countries carry.
- **Shrinkage:** Denmark ($n = 7$) moves from a raw 100% repurchase rate to 56.5%; Cyprus ($n = 5$) from 60% to 45.9%; one-customer countries move to about 38–46%. Larger countries are shrunk less but still move: France ($n = 59$) goes from 57.6% to 47.9%, because 59 customers is few next to the UK’s 3,593 (Figure 8.2).

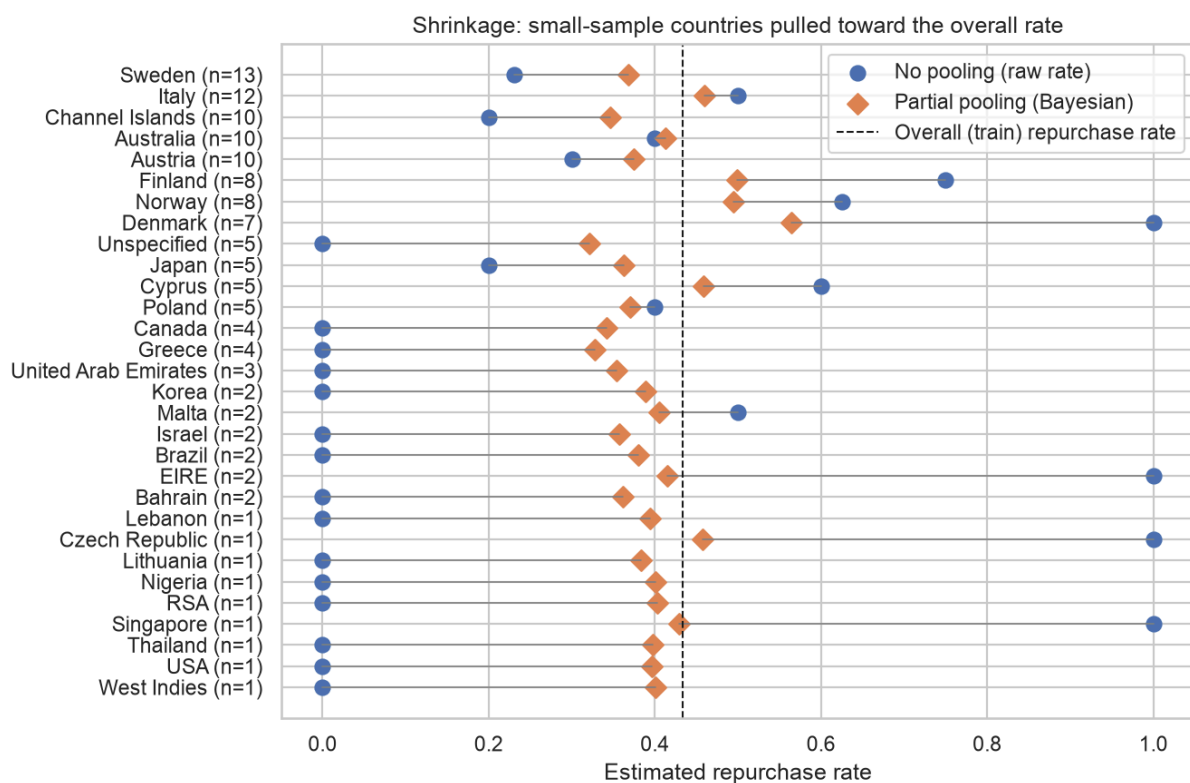


Figure 8.2: Raw (no-pooling) vs partially pooled repurchase rates for countries with fewer than 15 customers.

Prior sensitivity. The model was refitted with a tighter and a wider prior on sigma_a, using identical sampler settings:

Prior on sigma_a	sigma_a mean [5%, 95%]	Divergences	Denmark	France	USA (n = 1)
HalfNormal(0.5)	0.36 [0.04, 0.79]	0	52.5%	47.1%	40.6%
HalfNormal(1) (main)	0.46 [0.05, 1.00]	0	56.5%	47.9%	39.6%
HalfNormal(2.5)	0.52 [0.06, 1.11]	1	58.3%	48.2%	38.5%

The country estimates are **robust** to the prior choice, moving by a few points at most. The between-country spread itself is prior-sensitive, as expected given how little data inform it.

Predictive calibration (test set). Probabilities are averaged over all 8,000 posterior draws. The 3 test customers from countries never seen in training get a new country baseline drawn from the fitted distribution, something a no-pooling model cannot do at all.

Test set	AUC	Brier	Mean predicted (actual 0.434)
Hierarchical Bayes (2 predictors + country)	0.798	0.181	0.441
Logistic, 9 features (reference)	0.810	0.175	0.438

The hierarchical model is well calibrated. It ranks slightly worse only because it uses two predictors rather than nine.

Limits of the country estimates. For the 26 countries with fewer than 10 customers, the estimate is mostly the pooled average plus a small nudge from their own data. That is the correct behaviour, but it means the model **cannot reveal** that a small market is genuinely different. It can only stop small samples from producing extreme estimates. The estimates apply to identifiable customers only (Task 3), and the model includes only two behavioural predictors.

8.3 Bayesian decision making

Corrected rule. An earlier version computed “expected value” from each test customer’s **realised** future spend, which is unknown when the decision is made, and multiplied the offer cost by the repurchase probability. As a result the probability had no effect on the decision. The rule now uses only information available at the cutoff:

contact if the posterior expectation of **uplift × margin × P(repurchase) × E[spend | repurchase] – offer cost** is > 0,

with P(repurchase) taken from all 8,000 posterior draws and E[spend | repurchase] from OLS on log(spend) with Duan smearing, the conditional-spend model selected in Task 5 (a frequentist plug-in, stated as such).

- **Prospective result (placeholder assumptions: 30% margin, £10 offer, 10% uplift).** The rule contacts customers with expected 90-day spend above £333: **38.5% of test customers** (506 of 1,314), for a model-predicted incremental profit of **£12.6k (90% posterior interval £12.2k–£13.0k)**. The interval covers uncertainty in the repurchase probabilities only; it does not cover the spend prediction or the assumed uplift. For 18 of the contacted customers, the posterior probability that contacting them pays off is below 90%.
- **Retrospective check (not deployable).** On realised spend, the same list would have earned £11.8k, against £6.5k for contacting everyone. This validates the ranking, not the profit level.
- **The unknown that matters is the uplift.** Uplift and margin enter only as a product. Across plausible values the contact share runs from 4% to 84%, and predicted profit from £1.3k to £50k. P(repurchase) is a

prediction under historical conditions, not a no-offer counterfactual. The offer's causal effect needs a randomised experiment (Task 6).

8.4 Evaluation

	Assessment
Are Bayesian approaches appropriate?	Yes, for two specific jobs: estimating rates for thin groups (countries, new accounts), and making decisions that carry their uncertainty through. Not as a replacement for the Task 5 model on the bulk of the customer base
Advantages	Principled shrinkage for small groups; a full posterior for every quantity, so decisions can use “probability that this pays off” rather than a point estimate; a prediction for an unseen country; priors that make assumptions explicit and testable
Limitations	Priors can be contested (here the country estimates were checked and found robust, but σ_a is not); MCMC is computationally heavy (about a minute per fit here) and needed tuning to converge; it is harder to explain to non-technical managers than an odds ratio; Naive Bayes’s independence assumption is clearly violated
When Bayesian methods outperform traditional ones	Many small groups (26 countries with fewer than 10 customers); decisions where uncertainty matters (contact only if the probability of profit is high); entering a new market with no history; combining prior knowledge, such as an expert’s estimate of the offer effect, with data

Recommendation. Keep the Task 5 LASSO model (selected under the Brier rule) for ranking the main (UK-dominated) customer base. Use the hierarchical model’s pooled estimates for thin or new international markets. Use the Bayesian decision framework only once the uplift has been measured.

References (Task 8)

Full entries are in the Task 2 references (section 2.9): Abe (2009); Domingos & Pazzani (1997); Efron & Morris (1975); Gelman (2006); Niculescu-Mizil & Caruana (2005).

Task 9 – Time Series Analysis

Owner: Member D. The brief does not require time-series analysis, only a critical discussion. This project goes further and runs the diagnostics, because they turn the discussion's main caveat (too little history) into evidence. Results come from `notebooks/r/06_time_series.R` (weekly revenue, 104 weeks, November 2009 – December 2011).

Status: exploratory only. The fitted model performed **worse** than a naive benchmark. Nothing in this section supports deploying a forecasting model yet.

9.1 Trend analysis

Weekly revenue was aggregated from the cleaned invoice lines. Outside the fourth quarter, the two years look similar. Revenue rises from August/September, peaks in November (£1.43M in November 2010, £1.45M in November 2011, about 2.2 times a typical off-peak month), and falls sharply in December. The business effectively pauses over Christmas and New Year: revenue is £0 in the weeks ending 3 January 2010 and 2 January 2011, which is a structural break any model has to handle. Stationarity tests **disagree**: ADF ($p = 0.61$) cannot reject a unit root, while KPSS ($p = 0.091$) does not reject stationarity at the 5% level. `ndiffs()` suggests one order of differencing. The series sits in an unclear zone between trending and stationary.

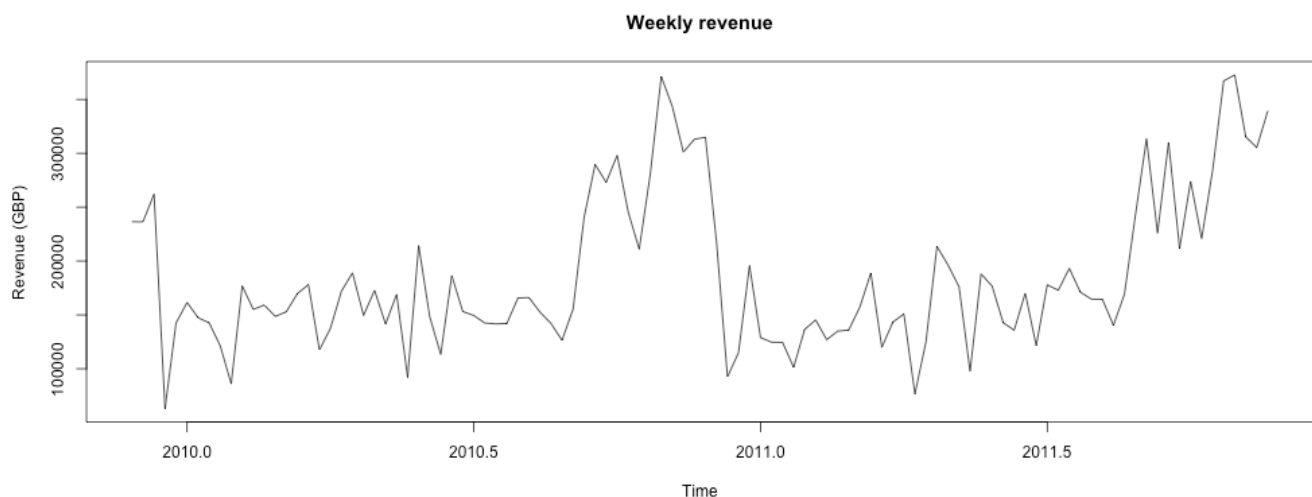


Figure 9.1: Weekly revenue, November 2009 – December 2011.

9.2 Seasonal analysis

- **Weekly pattern:** Saturday revenue is essentially zero (£9,803 in total, against £2.96M–£4.01M for each weekday) and Sunday is well below weekday levels. This is a B2B trading rhythm.
- **Annual pattern:** the data contain **only about two annual seasonal cycles**. Classical decomposition reports seasonal and trend strength of essentially 1.0, but that is an artefact, not evidence of strong seasonality. With a 52-week period, each week-of-year's seasonal value is estimated from at most **two** observations, so the decomposition fits almost perfectly by construction. **The seasonal strength estimates are unstable and should not be read at face value.**

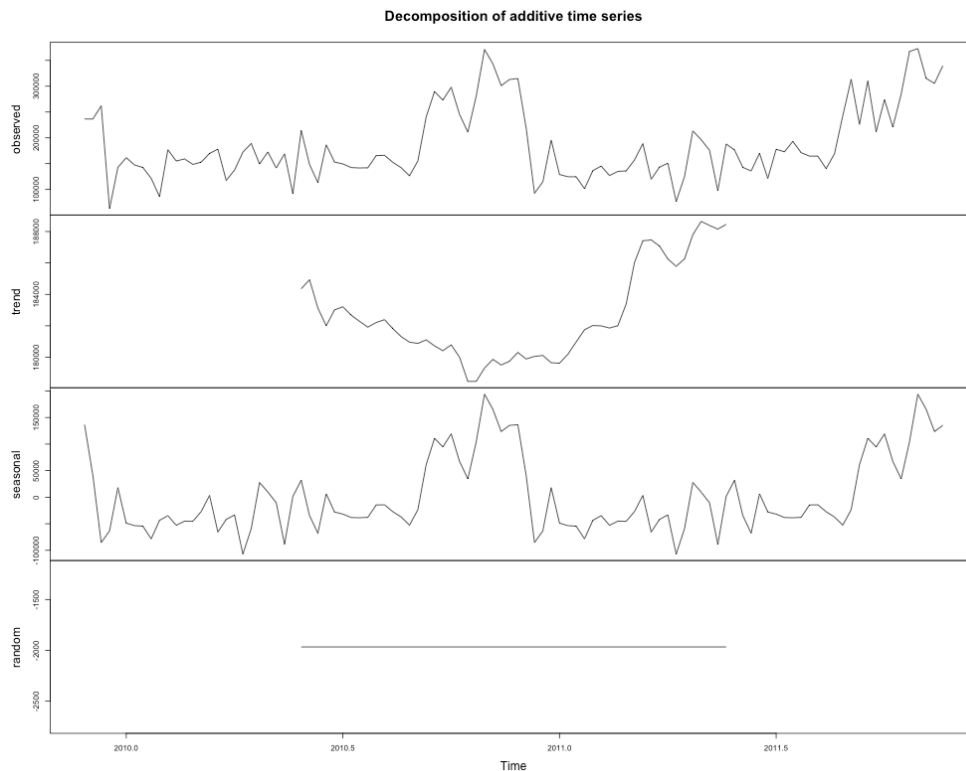


Figure 9.2: Classical decomposition of weekly revenue. The near-perfect seasonal fit reflects only two cycles of data.

9.3 Forecasting and ARIMA modelling

The last 8 weeks were held out as a test period, and `auto.arima()` (Hyndman & Khandakar, 2008) was fitted on the remaining 96 weeks.

- It selected a plain, **non-seasonal ARIMA(1,0,0)**. It could not attempt a seasonal (SARIMA) model, because 96 weeks is less than two full 52-week cycles.
- Its residuals show no remaining autocorrelation (Ljung–Box $p = 0.69$), so the model is not mis-specified for the little structure it can capture.
- **Benchmark comparison.** On the 8 held-out weeks, the model's RMSE was **£135,207**. Simply repeating the last observed week (a naive forecast) gave **£109,032**. **The fitted model performed worse than the naive benchmark.**

This agrees with forecasting research: simple benchmarks are hard to beat, and any model has to be compared against them before use (Fildes et al., 2022; Makridakis et al., 2022). Three independent symptoms point to the same root cause: the suspiciously perfect decomposition, the conflicting stationarity tests, and `auto.arima`'s inability to fit a seasonal term. That root cause is **too little history**.

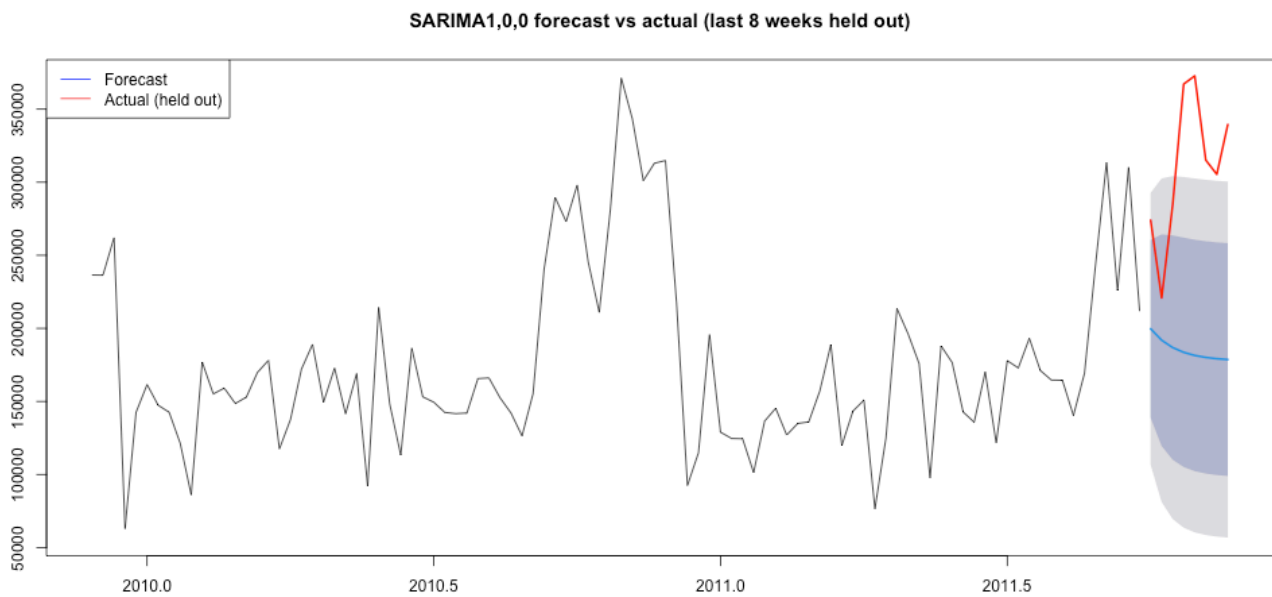


Figure 9.3: ARIMA forecast for the 8 held-out weeks against actual revenue.

9.4 How time series analysis should be incorporated

With **at least three to four years** of weekly data, a seasonal model (SARIMA, or exponential smoothing with a seasonal term) could be estimated properly and validated over a full held-out season. Until then:

- Plan Q4 from the **calendar and last year's pattern plus a buffer**, not from a fitted model (Task 12, O3 and R4).
- Keep the **naive and seasonal-naive forecasts** as the benchmark any future model must beat.
- Handle the Christmas shutdown weeks explicitly as a structural break.

9.5 Business applications and expected benefits

Application	How it would be used	Expected benefit (once enough history exists)
Inventory planning	Forecast the Q4 build-up, which starts in August/September	Less stock-out risk at the peak, less over-stock afterwards
Staffing and dispatch	Weekday rhythm, no Saturday trading, 10:00–16:00 order peak	Capacity matched to demand
Campaign timing	Retention campaigns (Task 10) timed before the peak and the Christmas shutdown	Offers land when customers are planning orders
Cash-flow planning	Expected revenue path through the year	Earlier view of working-capital needs ahead of Q4

The descriptive pattern is already actionable for planning. A forecasting **model** is not yet justified.

References (Task 9)

Fildes, R., Ma, S., & Kolassa, S. (2022). Retail forecasting: Research and practice.

International Journal of Forecasting, 38

(4), 1283–1318. <https://doi.org/10.1016/j.ijforecast.2019.06.004>

Hyndman, R. J., & Khandakar, Y. (2008). Automatic time series forecasting: The forecast package for R.

(3), 1–22. <https://doi.org/10.18637/jss.v027.i03>

Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2022). M5 accuracy competition: Results, findings, and conclusions.

(4), 1346–1364. <https://doi.org/10.1016/j.ijforecast.2021.11.013>

Task 10 – Industry Innovation Proposal: a Customer Repurchase Early-Warning System

Owner: Member D. Every number is taken from the project's analysis notebooks, cited in brackets. Proposed thresholds, roles and timelines are proposals to confirm with the client, not facts about the organisation. Costs, margins and offer effects are not known and are marked as assumptions wherever they appear.

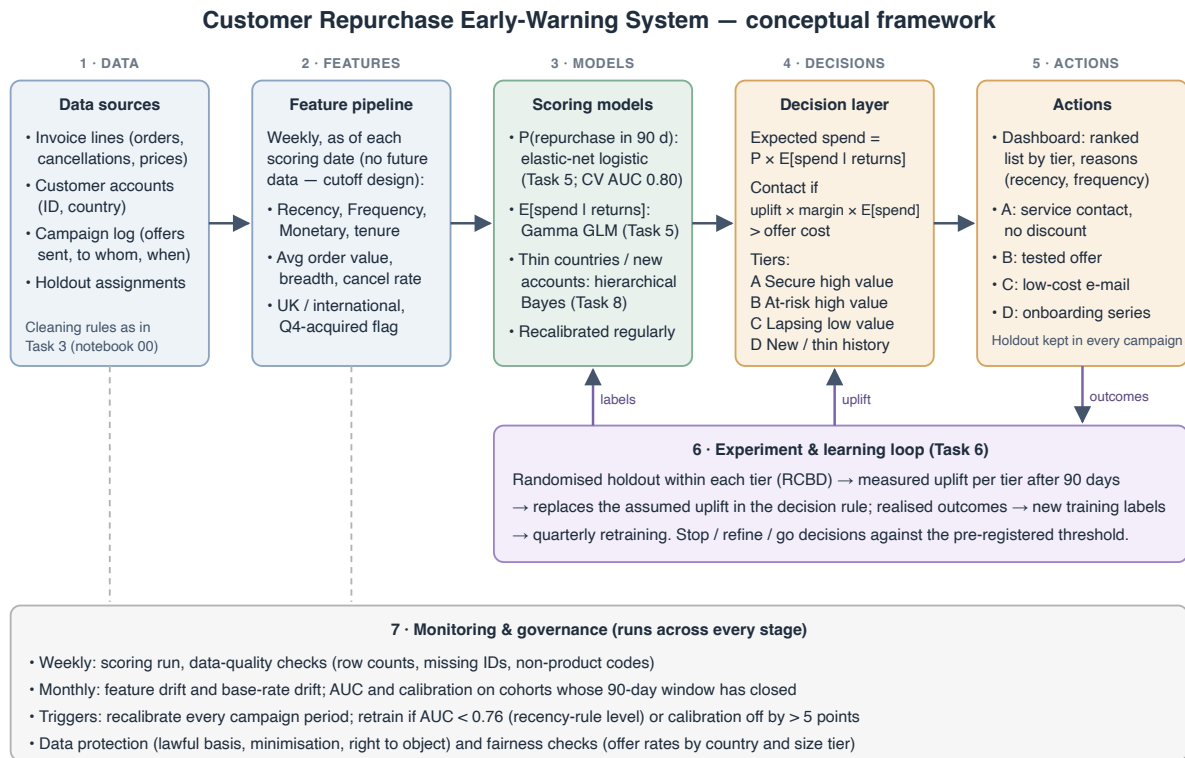


Figure 10.1: Conceptual framework. Stages 1–5 run every week. Stage 6 (the experiment loop) supplies the one number the models cannot: the offer's real effect. Stage 7 governs all stages.

10.1 Business need

- **Most of the customer base is inactive at any moment.** Only 43.4% of customers bought again in the 90 days after the cutoff. 30% have placed a single order, and the median customer last ordered 162 days before the cutoff (notebook 01).
- **Revenue depends on a few accounts.** The top 1% of customers bring in 30.8% of revenue and the top 10% bring in 62.7% (notebook 01). Losing a handful of large wholesale buyers would hurt materially, and a lapse is currently noticed only after the fact.
- **The signal exists before the lapse.** Past behaviour alone ranks customers well: repeated cross-validation AUC 0.799 (SD 0.015), with the top 10% of scores repurchasing at about twice the base rate (lift 2.09), well above the “most recent customers first” rule (AUC 0.762) (notebook 03, A9). Nobody at the retailer uses that signal today.
- **Spending on retention is currently unguided.** There is no evidence yet on whether offers work, or for whom (Task 6).

Business objective. Identify, every week, which customers are drifting away and which of them are worth acting on, so that retention effort goes where it produces *measured* incremental margin. The target is to raise the 90-

day repurchase rate among contacted at-risk, high-value accounts above that of a randomised holdout, by at least the pre-registered threshold from the Task 6 pilot.

10.2 Proposed solution

A weekly scoring and decision pipeline with a built-in experiment loop (Figure 10.1).

Component	Design	Grounded in
Data inputs	Invoice lines (orders, cancellations, prices, dates), customer accounts (ID, country), and a new campaign log recording who was sent which offer and when, plus holdout assignments	Task 3 data; the campaign log is new and needed for Stage 6
Feature pipeline	Weekly job that recomputes, as of the scoring date only, recency, frequency, monetary value, tenure, average order value, product breadth, cancellation rate, UK/international and Q4-acquired flag. Uses the same cleaning rules as notebook 00 (duplicates, non-product codes, non-positive prices, reversed orders)	The cutoff design (<code>src/features.py</code>) guarantees no future data leaks into a score
Model outputs	Per customer: P(repurchase in the next 90 days) from the LASSO logistic model (selected under the Task 5 Brier rule); E[spend if they return] from log-OLS with Duan smearing; expected spend = the product. Used for ranking : the probability and £ levels shift between seasons. Countries or accounts with very little history get a pooled estimate from the hierarchical Bayesian model	Tasks 5 and 8
Decision thresholds	Contact if $\text{uplift} \times \text{margin} \times \text{expected spend} > \text{offer cost}$. With placeholder values (10% uplift, 30% margin, £10 offer) that means expected spend > £333, which would select about 44% of customers (notebook 03, C3). The uplift must come from the pilot , not from assumption	Task 5, C3; Task 6
Dashboard tiers	A. Secure, high value: high P and high expected spend. B. At risk, high value: high expected spend but P falling as recency grows. C. Lapsing, low value: low P and low expected spend. D. New / thin history: tenure under 90 days or an unseen country, scored with pooled estimates	Tier cut-offs set from score quantiles and reviewed after the pilot
Campaign actions	A: account-manager service contact, <i>no discount</i> . The top tier repurchases at about 81% anyway (notebook 07), so a discount mostly rewards purchases that would have happened. B: the offer tested in the pilot, if it passed. C: low-cost automated e-mail only. D: onboarding sequence	Ascarza (2018); Devriendt et al. (2021)
A/B feedback loop	Every campaign keeps a randomised holdout within each tier (the RCBd from Task 6). After 90 days, the measured uplift per tier replaces the assumed uplift in the decision rule, and realised outcomes become new training labels	Kohavi et al. (2009); Lemmens & Gupta (2020)

The dashboard shows each account's tier, its score, and the plain-language reasons behind it (e.g. “no order for 140 days; ordered monthly before”). Account managers get a reason, not just a number.

10.3 Expected benefits

- **Earlier action on the accounts that matter.** A lapsing top-decile account is flagged weeks before anyone would currently notice.
- **Retention budget guided by evidence, not by guesswork.** The experiment loop turns the assumed uplift, currently the largest uncertainty in the profit estimate (contact share 4%–82% depending on it, notebook 03 C3), into a measured one.
- **Less money wasted on customers who would buy anyway.** Secure accounts get service, not discounts.
- **One consistent definition of an “active customer”** across sales, marketing and finance.

No £ benefit is claimed. The analysis cannot estimate one until the pilot measures the offer's effect (Task 6). Any figure quoted before then would be an assumption presented as a result.

10.4 Implementation plan

Phase	Duration (proposed)	What happens	Exit criterion
0. Data foundation	4 weeks	Automate the weekly extract; port the notebook 00 cleaning rules into a scheduled job; create the campaign log	Weekly customer table matches a manual rebuild
1. Shadow scoring	6 weeks	Score weekly and publish the dashboard, no campaigns . Check that scores are stable and that tiers make sense to account managers	Account managers agree the tiers are usable; AUC on the first matured cohort ≥ 0.76
2. Pilot	about 4 months (2 weeks set-up + 90-day window + read-out)	Task 6 RCBD pilot, launched outside October–December	Pre-registered stop / refine / go decision
3. Rollout with holdout	ongoing	Tiered campaigns, a permanent randomised holdout (e.g. 10–20% of each tier), quarterly retraining	Quarterly KPI review

Required systems: a scheduled data extract from the order system; a scripting environment for the pipeline (the project's Python code is the prototype); a dashboard tool; the existing e-mail platform, able to send unique single-use codes.

Required staff (proposed): a *responsible owner* on the business side (e.g. the head of sales or customer retention; to be confirmed with the client) who decides campaigns and owns the KPIs; a data analyst (part-time) who runs and monitors the pipeline; a marketing executive who runs the campaigns; account managers who act on tiers A and B; and the data protection lead for sign-off.

10.5 KPIs and monitoring

Level	KPI	Proposed target / trigger	Why this number
Model	AUC on cohorts whose 90-day window has closed	Retrain if it falls below 0.76	0.76 is what the simple recency rule achieves (notebook 03, A9); below it, the model adds nothing
Model	Calibration: mean predicted vs actual repurchase rate	Alert if off by more than 5 points ; any recalibration must be validated before use (a simple previous-period correction made the next period worse in Task 5)	Out-of-time tests showed the probability level shifts by 12–15 points between seasons while the ranking holds (notebook 03, A8); miscalibrated probabilities mislead decisions (Van Calster et al., 2019)
Data	Feature drift and base-rate drift	Monthly check against the training period	Base rates ranged from 32% to 58% across the cutoffs tested (notebook 03, A8)
Data	Data-quality checks	Weekly: row counts, share of missing customer IDs (22.8% in the raw data), non-product codes	Task 3 cleaning audit
Business	Incremental repurchase rate, treated vs holdout, per tier	At least the pilot's pre-registered threshold	The only KPI that measures the <i>effect</i> of the system
Business	Incremental margin per £ of offer cost	> 1	Break-even
Business	Unsubscribes and complaints per campaign	Agreed with marketing before launch	Guards against over-contacting

Monitoring cadence: scoring and data-quality checks weekly; drift and performance monthly; retraining and KPI review quarterly. Outcomes only become known 90 days after each score, so model performance is always measured on cohorts at least one quarter old. The system has a built-in lag, and that lag has to be planned for. Churn models have been shown to hold most of their accuracy when applied three months later (Neslin et al., 2006), which supports quarterly rather than monthly retraining; the out-of-time results here agree for ranking (AUC 0.80 vs 0.81) but not for calibration.

Model-drift controls: out-of-time validation before any retrained model replaces the current one; the recency rule kept as a permanently running benchmark; a rollback to the previous model version if the new one underperforms on the latest matured cohort.

10.6 Implementation challenges

- **Outcome lag.** It takes 90 days to learn whether a score was right, so problems surface slowly.
- **Seasonality.** Probability levels shift between seasons (Task 5), and a simple previous-period correction did not fix it, so calibration is a standing problem, not a one-off.
- **Missing identities.** 22.8% of raw invoice lines have no customer ID, and they are not missing at random (Task 3). The system can only see and act on identifiable accounts. Encouraging account log-in at checkout widens its coverage.
- **Thin international data.** 26 of 41 countries have fewer than 10 customers (notebook 05). Their estimates rely on pooling and should be shown with that caveat.
- **Organisational adoption.** Account managers must trust the tiers. Phase 1 (shadow scoring) exists to build that trust before any money is spent.

10.7 Data protection and fairness safeguards

- **Minimisation:** the pipeline uses only order behaviour and country. It needs no personal characteristics.
- **Lawful basis and objection:** the data protection lead confirms the lawful basis for profiling customers for marketing, and honours objections to direct-marketing profiling, particularly for accounts held by individuals such as sole traders.
- **Fairness monitoring:** offer rates are reported by country and customer-size tier each quarter. Whoever owns the system reviews them, so that systematically fewer offers to small or international customers is a deliberate policy, not an unexamined side-effect. Country enters the model only as UK vs international, plus pooled estimates for thin groups.
- **Transparency:** customers who receive the same offer get the same terms, and the reasons shown on the dashboard are behavioural and explainable.
- **Retention of experiment data:** holdout assignments are kept only as long as needed to analyse the experiment.

10.8 Organisational impact

The system changes retention from a reactive activity (noticing that a big customer has stopped ordering) into a weekly, measured routine. It gives sales, marketing and finance one shared view of customer health. It also builds the organisation's ability to test before spending, which transfers to pricing, product and channel decisions beyond retention. The time-series evidence (Task 9) supports running this system rather than a revenue-forecasting tool first. Two years of history is not enough for a reliable seasonal forecast (the fitted model lost to a naive forecast), while customer-level scoring works now.

References (Task 10)

Ascarza, E. (2018). Retention futility: Targeting high-risk customers might be ineffective.

Journal of Marketing Research, 55

(1), 80–98. <https://doi.org/10.1509/jmr.16.0163>

Devriendt, F., Berrevoets, J., & Verbeke, W. (2021). Why you should stop predicting customer churn and start using uplift models.

Information Sciences, 548

, 497–515. <https://doi.org/10.1016/j.ins.2019.12.075>

Kohavi, R., Longbotham, R., Sommerfield, D., & Henne, R. M. (2009). Controlled experiments on the web: Survey and practical guide.

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(1), 140–181. <https://doi.org/10.1007/s10618-008-0114-1>

Lemmens, A., & Gupta, S. (2020). Managing churn to maximize profits.

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(5), 956–973. <https://doi.org/10.1287/mksc.2020.1229>

Neslin, S. A., Gupta, S., Kamakura, W., Lu, J., & Mason, C. H. (2006). Defection detection: Measuring and understanding the predictive accuracy of customer churn models.

Journal of Marketing Research, 43

(2), 204–211. <https://doi.org/10.1509/jmkr.43.2.204>

Van Calster, B., McLernon, D. J., van Smeden, M., Wynants, L., & Steyerberg, E. W. (2019). Calibration: The Achilles heel of predictive analytics.

BMC Medicine, 17

, 230. <https://doi.org/10.1186/s12916-019-1466-7>

Task 11 – Industry Expert Validation

STATUS: NOT YET COMPLETED. No industry expert has been consulted yet, and this section contains **no expert feedback**. Everything below is a plan, an interview guide and empty evidence templates. Task 11 must not be presented as complete, and no recommendation in Task 12 may cite expert support, until the consultation has taken place and documentary evidence (interview summary, meeting minutes, e-mails, meeting screenshots or a completed feedback form) is stored in [docs/evidence/](#) and referenced here.

Owner: Member D (with the whole team for the consultation itself).

11.1 What the brief requires

At least one industry expert relevant to the domain must give professional feedback on: the **relevance of the problem**, the **practicality of the recommendations**, **implementation feasibility** and **organisational impact**. The report must then show **how the feedback changed the final recommendations**.

11.2 Who to consult

Any **one** of the following profiles is relevant (in order of preference):

1. A CRM, customer-retention or e-commerce manager at a wholesale or online retail business.
2. A sales or account manager who manages B2B trade customers (the retailer's customers are largely wholesalers).
3. A marketing-analytics or data-science practitioner who has run retention campaigns or A/B tests in retail.

Record the expert's name (or an anonymised title if they prefer), organisation type, role, years of experience, and why they are relevant. The consultation also needs their consent to be named and quoted.

11.3 Why the expert matters to *this* analysis

Several numbers in the analysis are **placeholders** that only someone with industry knowledge can replace. The expert consultation is the planned source for them:

Assumption in the analysis	Current placeholder	Where it is used	What it changes
Gross margin on revenue	30%	Notebooks 03 (A5, C3) and 05; Task 6 break-even	Contact threshold, predicted profit
Cost of one retention offer (voucher + handling)	£10	Same	Contact threshold, break-even uplift
Plausible incremental effect of an offer	10% of expected spend	Same	Contact share ranges from 4% to 82% depending on it (notebook 03, C3)
Tolerable control-group size for a pilot	20–50%	Task 6	Detectable effect: 6.1 vs 3.9 points
Lead time needed before the Q4 peak	Build-up from August / September (notebook 01)	Task 12 timing	Pilot and campaign calendar
How B2B customers are contacted today	Unknown	Task 10 actions	Whether e-mail offers or account-manager calls are realistic

11.4 Interview guide (about 30–40 minutes)

Before the meeting, send a two-page summary of the findings (Tasks 3–5 headline results, the Task 6 pilot design and the Task 10 framework). Ask the questions below and record answers **in the expert's own words**.

A. Relevance of the problem

1. How does a business like this currently notice that a trade customer has stopped ordering?
2. Is “no order in the next 90 days” a sensible definition of a lapsed customer in wholesale giftware, or would you use a different window?
3. Our data show that the top 10% of customers bring in about 63% of revenue. Does that match your experience, and how is that dependence managed in practice?

B. Practicality of recommendations

4. What gross margin and what cost per retention offer would be realistic for a UK online gift wholesaler?
5. From your experience, what size of effect does a retention offer typically have on repeat ordering?
6. Would you give high-value, loyal accounts a discount, or a service contact instead?

C. Implementation feasibility

7. Could a business of this size keep a randomised control group that receives no offer for 90 days? What share would be acceptable?
8. Who would own a weekly at-risk customer list, and how would account managers want to receive it?
9. What data would realistically be available (order system, e-mail platform, campaign history)?

D. Organisational impact

10. Which of our recommendations would you prioritise, and which would you drop?
11. What risks or ethical concerns do you see (e.g. discount expectations, fairness to small customers, data protection)?
12. When should Q4 preparation start for a gift wholesaler?

11.5 Evidence log (to be completed)

Field	Entry
Expert role and relevance	<i>pending</i>
Organisation type	<i>pending</i>
Date of consultation	<i>pending</i>
Method (in person / video call / e-mail / feedback form)	<i>pending</i>
Team members present	<i>pending</i>
Consent to be named / quoted	<i>pending</i>
Evidence file(s) in docs/evidence/	<i>pending</i>

11.6 Findings (to be completed from the evidence)

Area	Expert's feedback (summarised, with quotes)	Practical constraint identified
Relevance of the problem	<i>pending</i>	<i>pending</i>
Practicality of recommendations	<i>pending</i>	<i>pending</i>
Implementation feasibility	<i>pending</i>	<i>pending</i>

Area	Expert's feedback (summarised, with quotes)	Practical constraint identified
Organisational impact	<i>pending</i>	<i>pending</i>

11.7 How the feedback changed the recommendations *(to be completed)*

Recommendation (Task 12 ID)	Before the consultation	After the consultation	Reason
<i>pending</i>			

After the consultation: replace the placeholders in 11.3 with the expert's figures, rerun notebooks 03 and 05 (section C3 and the decision rule) with those values, update the Task 6 break-even, and fill in the "Expert feedback" column of the Task 12 recommendation table.

11.8 Completion checklist

- ☐ Expert identified and agrees to participate (and to be named or anonymised)
- ☐ Findings summary sent in advance
- ☐ Consultation held; notes or recording taken
- ☐ Evidence saved in docs/evidence/ (summary, minutes, e-mails, screenshots or feedback form)
- ☐ Evidence log (11.5) and findings (11.6) completed
- ☐ Placeholder assumptions replaced and notebooks 03 and 05 rerun
- ☐ Change log (11.7) completed and Task 12 "Expert feedback" column filled in
- ☐ Status banner at the top of this file removed

Task 12 – Final Consultancy Recommendations

Owner: all members. Every recommendation cites the statistical result it rests on (with the notebook or script it came from), at least one published source, and a column for industry expert feedback.

Expert feedback is pending. No expert has been consulted yet (Task 11 is incomplete), so the “Expert feedback” column reads *Pending* throughout. These recommendations are **provisional** until that column is filled in, and several of the quantities below (margin, offer cost, uplift) are placeholders that the consultation is meant to replace.

12.1 Summary for senior management

1. **Do not fund blanket retention discounts yet.** The models reliably identify who is likely to buy again, but not whether an offer changes that. Depending on the offer’s unknown effect, the profitable contact share ranges from 4% to 82% of customers.
2. **Run a four-month randomised pilot first,** launched outside the Q4 peak, with repurchase as the success measure and pre-agreed stop/go rules.
3. **Build the weekly early-warning system in stages.** Start by scoring customers and publishing the dashboard without running campaigns, and only run campaigns once the pilot has measured the offer’s effect.
4. **Protect the top accounts.** The top 10% of customers bring in 63% of revenue. Give them service contact, not discounts, because 81% of the most loyal tier buys again anyway.
5. **Plan Q4 from the calendar, not from a forecast model.** Two years of data cannot yet support a reliable seasonal forecast.

12.2 Key quantities

Decision parameter	Recommendation	Basis
Targeting tier for offers	Tier B (at risk, high value): expected spend above the economic threshold. With placeholder values the threshold is expected 90-day spend > £333, which selects about 44% of customers. Recompute it with the pilot’s measured uplift	Notebook 03, C3
Campaign size in the pilot	The whole active base (about 5,250 accounts), 50/50 within each block (detects +3.9 points), or 80/20 if a large control is unacceptable (detects +6.1 points)	Notebook 07
Pilot duration	About 4 months: 2 weeks set-up, 90-day outcome window, 2 weeks analysis	Task 6
Pilot timing	Launch outside October–December (e.g. March, read out in June); repeat before Q4 if it succeeds	Notebook 03 A8: base rates differ by season (32% vs 43%)
Success threshold	Repurchase uplift ≥ 4 points, 95% CI excluding 0, and incremental margin > offer cost using the confirmed margin	Notebook 07; Task 6 break-even of 2.7–6.3 points
Stop / go	Go: roll out tiered offers. Refine: effect positive but below threshold, so test a cheaper or better-targeted offer. Stop: CI includes 0, so no blanket offers	Task 6
Review frequency	Score weekly; monitor drift and performance monthly; retrain quarterly; retrain early if AUC < 0.76; recalibrate only with a validated method	Notebooks 03 and 03b
Q4 preparation period	Plan stock and staffing in June–July and build from August, because revenue rises from August/September and peaks in November at about 2.2× a typical off-peak month	Notebook 01

12.3 Recommendations and evidence

Each row lists the recommendation, the statistical evidence behind it, the literature that supports it, and the expert feedback, which is pending for every row until Task 11 is completed.

Strategic

ID	Recommendation	Statistical evidence	Literature	Expert feedback
S1	Do not fund blanket retention discounts until a randomised pilot has measured the offer's effect	The models predict repurchase under historical conditions, not response to an offer. The prospective profit of targeting depends almost entirely on the assumed uplift: contact share 4%–82% and predicted profit £1.4k–£47k across uplift × margin 1%–8% and costs of £5–£20 (notebook 03, C3; notebook 05 gives the same pattern)	Ascarza (2018); Lemmens & Gupta (2020); Devriendt et al. (2021)	<i>Pending (Task 11)</i>
S2	Run the RCBD retention pilot in Task 6 (parameters in 12.2)	Repurchase is detectable: +5 points needs 1,558 per arm (CRD) or about 1,193 (RCBD). Spend is not: +10% needs about 55,000 per arm. Recency × frequency blocks explain 23% of repurchase variance (notebook 07)	Kohavi et al. (2009); Montgomery (2017)	<i>Pending (Task 11)</i>
S3	Adopt the early-warning system in Task 10, starting with 6 weeks of shadow scoring	Past behaviour ranks customers well and consistently: repeated CV AUC 0.799 (SD 0.015), top-decile lift 2.09, versus 0.762 and 1.77 for the recency rule (notebook 03, A9). Ranking survives out-of-time testing (AUC 0.797–0.800, notebook 03, A8)	Neslin et al. (2006); Tamaddoni Jahromi et al. (2014); Chou et al. (2022)	<i>Pending (Task 11)</i>
S4	Treat the top accounts as a managed portfolio: service contact, not discounts	Top 1% = 30.8% of revenue; top 10% = 62.7% (notebook 01). The most loyal recency × frequency tier repurchases at 81% without any offer (notebook 07)	Ascarza (2018); Gattermann-Itschert & Thonemann (2022)	<i>Pending (Task 11)</i>

Operational

ID	Recommendation	Statistical evidence	Literature	Expert feedback
O1	Score customers with the LASSO logistic model, selected under a Brier-score rule declared before the results; treat Ridge as an equally defensible alternative	Mean development Brier over four rolling-origin periods: LASSO 0.2026, Ridge 0.2027 (difference +0.00001, 95% CI [−0.00014, +0.00016]); all candidates tie on AUC to two decimals. Locked and evaluated on Sep–Dec 2011: Brier 0.2005 vs 0.2014 for the full logistic (difference −0.0009 [−0.0012, −0.0005]) (notebook 03b)	Tibshirani (1996); Hoerl & Kennard (1970); Van Calster et al. (2019)	<i>Pending (Task 11)</i>
O2	Use model scores to rank customers, but do not use the raw probabilities or £ predictions as calibrated budget inputs across seasons until a recalibration method has been validated	Out-of-time AUC stays at 0.79, but the locked model predicts 0.30 against an actual 0.43 (calibration intercept +0.63); development intercepts ranged from −1.76 to +0.40, and a previous-period intercept correction made the evaluation Brier worse (0.2005 → 0.2063) (notebook 03b)	Van Calster et al. (2019)	<i>Pending (Task 11)</i>
O3	Prepare for Q4 from the calendar: plan in June–July, build stock and staffing from August	November revenue £1.43M (2010) and £1.45M (2011), about 2.2× the median off-peak month; the rise starts in August/September (notebook 01). Saturday revenue is essentially zero, so weekend staffing can stay minimal (script 06)	Fildes et al. (2022)	<i>Pending (Task 11)</i>

ID	Recommendation	Statistical evidence	Literature	Expert feedback
O4	Manage international accounts as a separate segment with order-size-aware account management	International customers' typical order value is 1.68× UK's, 95% CI [1.55, 1.82] (Test 1). It is also far more variable: variance ratio 5.24, bootstrap 95% CI [1.94, 13.39] (Test 3) (script 02)	Gattermann-Itschert & Thonemann (2022); Delacre et al. (2017)	<i>Pending (Task 11)</i>
O5	Use “acquired in Q4” as one scoring input, not as a stand-alone targeting lever	Q4-acquired customers repurchase 9.6 points more, 95% CI [6.7, 12.5], but Cramér's V is only 0.09, and the effect may be seasonal rhythm rather than acquisition timing (Test 2, script 02)	Lin et al. (2013); Sullivan & Feinn (2012)	<i>Pending (Task 11)</i>

Risk management

ID	Recommendation	Statistical evidence	Literature	Expert feedback
R1	Treat all £ profit figures as scenarios. Do not set a retention budget from them	Realised-spend profits (e.g. £9.3k for the top 10%) are retrospective and use information not available at decision time. Prospective figures depend on an assumed uplift (notebook 03, A5 and C3; notebook 05)	Verbeke et al. (2012); Lemmens & Gupta (2020)	<i>Pending (Task 11)</i>
R2	State plainly that the system covers only identifiable customers, and increase coverage (e.g. encourage account log-in)	22.8% of raw lines (13.6% of revenue) have no customer ID. The missingness is not random: a model predicts it with AUC 0.80 (notebooks 00, 01)	Rubin (1976); Little (1988)	<i>Pending (Task 11)</i>
R3	Use pooled (hierarchical) estimates for thin countries and show them with their uncertainty	26 of 41 countries have fewer than 10 customers. Pooled estimates are stable across priors (Denmark 52.5%–58.3%), but the between-country spread is uncertain (sigma_a 89% interval 0.06–0.99) (notebook 05)	Efron & Morris (1975); Gelman (2006)	<i>Pending (Task 11)</i>
R4	Do not deploy a revenue-forecasting model yet; use last year's pattern plus a buffer	With two seasonal cycles, auto.arima cannot fit a seasonal model, and its held-out RMSE (£135k) is worse than a naive forecast (£109k) (script 06)	Hyndman & Khandakar (2008); Makridakis et al. (2022)	<i>Pending (Task 11)</i>
R5	Monitor model drift monthly and keep the recency rule running as a benchmark	Base rates ranged from 32% to 58% across the cutoffs tested; the ranking held but calibration did not (notebook 03, A8)	Neslin et al. (2006)	<i>Pending (Task 11)</i>

Ethical

ID	Recommendation	Statistical evidence	Literature	Expert feedback
E1	Make the targeting of better prices a documented policy, and report offer rates by country and customer size each quarter	Value-based targeting concentrates offers: mean 90-day spend ranges from £74 to £1,732 across recency × frequency tiers (notebook 07), so small customers would systematically receive fewer offers	<i>No source in the current review; to be added</i>	<i>Pending (Task 11)</i>
E2	Limit experiment control periods to 90 days, use the same voucher terms for all recipients, and do not re-assign the same customers to control repeatedly	Pilot design (Task 6)	Kohavi et al. (2009)	<i>Pending (Task 11)</i>

ID	Recommendation	Statistical evidence	Literature	Expert feedback
E3	Confirm the lawful basis for profiling, keep data use to order behaviour and country, and honour objections to marketing profiling	The models need no personal characteristics: all nine features are behavioural or geographic (notebook 03)	<i>No source in the current review; to be added (e.g. the UK ICO's guidance on profiling for direct marketing)</i>	<i>Pending (Task 11)</i>

Future research

ID	Recommendation	Statistical evidence	Literature	Expert feedback
F1	Once pilot data exist, build uplift models that predict who responds to an offer, not who repurchases	The current models answer the wrong question for targeting (S1)	Devriendt et al. (2021)	<i>Pending (Task 11)</i>
F2	Add probabilistic “buy till you die” predictions as features, and test season-aware calibration	The selected expected-spend system under-predicts the Q4 level (mean £294 vs £530) and the probability level shifts by season (notebook 03b)	Fader et al. (2005b); Chou et al. (2022)	<i>Pending (Task 11)</i>
F3	Collect at least 3–4 years of weekly history before relying on seasonal forecasts	Two cycles give a near-perfect but meaningless seasonal fit and conflicting stationarity tests (script 06)	Hyndman & Khandakar (2008); Fildes et al. (2022)	<i>Pending (Task 11)</i>
F4	Explore product-theme segments (binary customer × product SVD) for product-specific offers	The themes are interpretable (e.g. home décor vs children's items). They add AUC +0.0057, 95% CI [+0.0002, +0.0111]: small for prediction, more useful for segmentation (notebook 04)	Jolliffe & Cadima (2016)	<i>Pending (Task 11)</i>

12.4 What these recommendations do not claim

- No recommendation claims that offers **cause** repurchase. That is exactly what the pilot is for.
- No £ profit figure in this report is guaranteed or deployable.
- Customer-level conclusions apply only to account-holding customers with a recorded ID.
- Every recommendation above is provisional until the expert-feedback column is completed (Task 11).

References (Task 12)

All sources are listed in full in the Task 2 references (`docs/task1_task2_industry_and_research.md`, section 2.9), except Montgomery (2017), which is listed under Task 6.